

A Lifelong Hard Gainer's Road to Recovery & Hypertrophy

Complete Case File

Left-Chain Dysfunction, Reincorporation & Load-Transfer Investigation

Author: H.E.

Version: 1.0

Record Through: August 28, 2026

First Published: September 2026

Document Status: Initial Release

Copyright © 2026 H.E.

Permission is granted to download, read, print, and redistribute these publications in their complete, unmodified form for personal, educational, and research purposes.

Redistribution must include attribution to the author.

Commercial use, modification, translation, adaptation, or the creation or distribution of derivative works requires the prior written permission of the author.

Disclaimer

This publication documents a longitudinal self-case investigation conducted by the author. It is intended for educational, investigative, and discussion purposes. It is not medical advice, diagnosis, or treatment.

Suggested Citation

H.E. (2026). *A Lifelong Hard Gainer's Road to Recovery & Hypertrophy: Complete Case File — Left-Chain Dysfunction, Reincorporation & Load-Transfer Investigation* (Version 1.0).

Official Publication Repository

<https://github.com/HEDomain/lifelong-hard-gainer>

For the most current editions and official releases, visit the project's official GitHub repository.

Series Statement

This publication is one volume in the *A Lifelong Hard Gainer's Road to Recovery & Hypertrophy* series.

Edition Statement

This edition preserves the complete evidentiary record — including the hypotheses that didn't survive scrutiny and the models they were replaced by, not just the conclusions that came after them.

Table of Contents

Right-click and select "Update Field" (or press F9) to generate the table of contents.

Version History

Version	Date	Description
0.x	2026	Ongoing investigative notes, exercise logs, working models, and evidence collected during the active investigation.
1.0	August 28, 2026	First complete investigative archive assembled from the longitudinal record. Establishes the baseline case file for all future revisions.

Document Status

This document constitutes **Version 1.0** of *A Lifelong Hard Gainer's Road to Recovery & Hypertrophy: Complete Case File — Left-Chain Dysfunction, Reincorporation & Load-Transfer Investigation*.

It represents the first complete assembly of the investigative record and serves as the authoritative archive of observations, experiments, working models, and evidence accumulated through August 28, 2026.

The investigation remains active. Consequently, this document should not be interpreted as a final explanation of the condition under investigation. Rather, it captures the current state of knowledge at the time of publication.

Future discoveries, refinements, and corrections will be incorporated through documented version updates. Earlier findings will not be silently rewritten or replaced. Maintaining the historical progression of the investigation is considered essential to preserving the integrity of the evidentiary record.

Scope Statement

This case file documents a longitudinal self-case investigation undertaken to understand and correct a persistent pattern of left-sided functional dysfunction that interfered with movement quality, force production, resistance training, and long-term physical development.

The investigation was not designed to establish or challenge a medical diagnosis, nor was it intended to validate a predetermined theory. Its purpose was to observe, document, analyze, and iteratively refine an evidence-based understanding of the functional problem through repeated experimentation and longitudinal observation.

The scope of this document includes:

- The historical development of the investigation.
- Relevant personal, athletic, and medical background.

- Investigative methodology.
- Chronological reconstruction of significant events.
- Individual case records.
- Failed experiments and abandoned hypotheses.
- Exercise-specific evidence records.
- Evolution of the working model.
- The current multi-deficit working model.
- Outstanding uncertainties and future investigative priorities.

This document does **not** present treatment recommendations, establish clinical standards of care, or claim that the findings are generalizable beyond this individual case. All conclusions are limited to the evidence contained within the investigative record.

Definitions & Conventions

For the purposes of this investigation, the following conventions apply throughout the document:

- **Observation** refers to a directly perceived, repeatedly documented, or otherwise reproducible finding recorded during the investigation.
- **Interpretation** refers to the author's current explanation of one or more observations.
- **Working Model** refers to the best evidence-supported explanation available at a particular stage of the investigation. Working models are expected to evolve as additional evidence becomes available.
- **Hypothesis** refers to a proposed explanation or prediction requiring further testing.
- **Reincorporation** refers to the progressive return of meaningful participation by previously underperforming or poorly integrated regions within the larger movement system.
- **Load Acceptance** refers to the ability of a body region to receive and organize externally applied force while maintaining desired mechanical organization.
- **Load Transfer** refers to the effective transmission of accepted force through successive segments of the movement chain without undesirable mechanical breakdown.
- **Automaticity** refers to the stage at which previously deliberate movement strategies become consistently organized without conscious attention.

Unless explicitly stated otherwise, directional terminology (left, right, proximal, distal, anterior, posterior, superior, and inferior) refers to standard anatomical orientation.

Evidence Classification System

Every substantive statement within this case file is intended to fall into one of the following evidence categories:

Classification	Definition
Confirmed Medical Diagnosis	A diagnosis established by a qualified medical professional and documented within the investigative record.
Direct Observation	A repeatedly observed, personally experienced, or otherwise documented finding recorded during the investigation.
Working Interpretation / Working Hypothesis	The current explanation considered most consistent with the available evidence. These interpretations remain subject to revision.
Unknown	A question, mechanism, anatomical structure, causal relationship, or explanation that has not yet been established by the available evidence.

Maintaining the distinction between observed evidence and explanatory interpretation is a foundational principle of this investigation.

Working Model Definition

A **working model** is the best explanation supported by the available evidence at a particular stage of the investigation. It is neither presented as established fact nor as a medical diagnosis. Working models are expected to evolve as new observations, experiments, and evidence strengthen, refine, expand, or replace earlier explanations. Accordingly, previous working models are preserved within this document as part of the historical investigative record rather than retrospectively treated as incorrect.

Terminology Note

This investigation employs a consistent internal vocabulary developed to distinguish observations, investigative concepts, and functional relationships with greater precision than was considered possible using inconsistent everyday terminology alone.

Whenever possible, terminology remains stable throughout the document. If a term evolves or is redefined during the investigation, that evolution is documented rather than silently incorporated into later sections.

Readers are encouraged to interpret technical terms according to the definitions provided within this document rather than according to external usage.

SECTION 1 — INVESTIGATION OVERVIEW

1.1 Purpose of the Investigation

The purpose of this investigation was to understand and correct a persistent pattern of left-sided functional dysfunction that progressively interfered with movement quality, resistance training, force production, and long-term physical development.

The investigation was initiated after conventional approaches—strength training, repeated exercise exposure, rest, and general conditioning—failed to produce meaningful or lasting improvement. Rather than continuing to increase training volume or intensity without understanding the underlying problem, the investigation shifted toward identifying *why* normal training adaptations were not occurring.

The ultimate objective was practical rather than theoretical: to restore normal function sufficiently to pursue long-standing aesthetic fitness goals. Although the investigation produced broader observations regarding human movement, its primary purpose remained understanding and correcting one individual's persistent functional limitations.

1.2 Purpose of the Case File

The purpose of this Case File is to preserve the complete investigative history of a longitudinal self-case investigation into a persistent unilateral movement dysfunction. Rather than presenting only final conclusions, this document records the chronological progression of observations, hypotheses, experiments, revisions, and current working models that collectively produced the present understanding of the condition.

The Case File serves three primary functions.

First, it preserves the historical record of the investigation. Earlier observations, including those that were later revised or incorporated into broader explanatory models, remain intentionally documented to provide an accurate account of how the investigation evolved over time.

Second, it distinguishes observation from interpretation. Throughout the document, direct observations, exercise-specific findings, failed hypotheses, working models, and current explanatory frameworks are presented as separate forms of evidence. This organizational structure allows readers to evaluate both the underlying observations and the reasoning that connects them.

Third, it provides a transparent evidentiary foundation for future investigation. By documenting not only successful discoveries but also abandoned hypotheses, unresolved

questions, and competing explanations, the Case File is intended to facilitate subsequent refinement rather than present the current model as a final or immutable conclusion.

Accordingly, this document should be understood as an investigative record rather than a diagnostic guideline, treatment protocol, or generalized clinical recommendation. Its primary objective is to preserve evidence, document reasoning, and support future inquiry through comprehensive and transparent documentation.

1.3 How to Read This Case File

This Case File is organized as an investigative record rather than a conventional narrative. Readers will obtain the clearest understanding of the investigation by following the document in the intended sequence.

The investigation begins with the subject background, methodology, and chronological record. These sections establish the historical context of the investigation and identify the major investigative milestones before individual findings are examined in detail.

The Case File Records (CF) then document the progression of significant observations, discoveries, and investigative transitions in chronological order. Each Case File represents a discrete event or developmental milestone that materially influenced the direction of the investigation.

Exercise-Specific Evidence Records (ER) should be interpreted as experimental environments rather than exercise prescriptions. Their primary purpose is to document how individual movements generated observations, strengthened existing findings, confirmed previous discoveries, or facilitated functional integration. The evidentiary importance of an exercise should not be interpreted solely from its role within the current training program.

Failed Experiments (FE) preserve hypotheses and explanatory models that were revised, narrowed, or abandoned as additional evidence accumulated. These records are included intentionally to preserve the historical development of the investigation and to demonstrate how competing explanations were evaluated over time.

Working Models (WM) document the progressive evolution of the investigation's explanatory framework. Earlier models should not necessarily be interpreted as incorrect; many were subsequently incorporated into broader and more comprehensive explanations. The Model Lineage accompanying each Working Model provides a traceable record of these relationships.

The Current Multi-Deficit Systems Model represents the present explanatory framework supported by the available evidence. It should be interpreted as the current working explanation of the observed functional relationships rather than a definitive anatomical or medical diagnosis.

Finally, the Open Case Items section identifies questions that remain unresolved. These entries intentionally preserve the current limits of the investigation and provide direction for future observation, experimentation, and refinement.

Together, the Chronological Record, Case Files, Exercise Evidence Records, Failed Experiments, Working Models, and Current Multi-Deficit Systems Model form a cumulative evidentiary structure in which conclusions emerge progressively from documented observations rather than from retrospective interpretation alone. The document is therefore intended to be read as an evolving investigative record, with each section providing context for those that follow.

1.4 Nature of the Investigation

This document records a **longitudinal self-case investigation** conducted over an extended period through systematic observation, repeated experimentation, iterative hypothesis development, and continuous refinement of the working model.

Unlike a controlled laboratory study, the investigation developed organically in response to persistent functional problems encountered during everyday movement and resistance training. New observations frequently redirected subsequent experimentation, producing an investigative process that evolved alongside the evidence itself.

The investigation therefore reflects an adaptive observational methodology rather than a predetermined experimental protocol.

1.5 Primary Objective

The primary objective was to identify the functional mechanisms preventing normal participation of the left lower kinetic chain during movement and resistance exercise so that those limitations could be corrected.

More specifically, the investigation sought to determine:

- why the left side consistently failed to participate normally under load;
- why conventional strength training produced incomplete results;
- why improvements in one region often failed to restore the larger movement system;
- which interventions consistently improved function;
- and how individual findings related to one another within an increasingly comprehensive working model.

The objective was understanding before correction. Any functional improvements observed during the investigation were evaluated not only for their practical benefit but also for the information they contributed toward explaining the larger problem.

1.6 Investigative Philosophy

From its inception, the investigation was guided by a simple principle:

If an approach repeatedly fails to solve a problem, repeating the same approach without understanding the failure is unlikely to produce a different result.

When conventional training reached a plateau, increasing effort alone was not considered an adequate solution. Instead, the investigation shifted toward observation and analysis.

Throughout the investigation, several operational principles emerged:

- Observe before concluding.
- Separate observation from explanation.
- Allow evidence to revise the working model.
- Preserve earlier interpretations as part of the historical record rather than rewriting them retrospectively.
- Prefer repeatable observations over attractive explanations.
- Continue testing until findings either persist or fail to reproduce.

These principles became increasingly formalized as the investigation progressed and ultimately shaped the methodology documented in this report.

1.7 Investigative Constraints

The investigation was conducted under numerous practical limitations that influenced both the available evidence and the pace of progress.

These included:

- absence of a formal research environment;
- no dedicated laboratory equipment;
- no biomechanical instrumentation;
- limited access to diagnostic imaging for functional questions;
- no continuous supervision by sports medicine or movement specialists;
- periods of limited access to training resources affecting continuity;
- interruptions caused by employment demands;
- progressive medical complications occurring during portions of the investigation;
- and reliance upon repeated self-observation and longitudinal documentation.

Despite these constraints, the extended duration of the investigation allowed repeated observations to accumulate over time, providing opportunities to distinguish transient findings from persistent functional patterns.

1.8 Investigative Value

This investigation is presented as the documented experience of a single individual and should be interpreted within that scope.

Its value does not depend upon claiming universal applicability. Rather, it lies in the unusually detailed longitudinal record of how one persistent functional problem was investigated over time.

Potential value may exist for:

- individuals attempting to understand persistent movement limitations;
- strength athletes experiencing unexplained asymmetries;
- clinicians interested in detailed patient-generated longitudinal observations;
- rehabilitation professionals examining the evolution of functional hypotheses;
- movement specialists interested in load acceptance, force transfer, and movement organization;
- researchers studying complex self-directed investigations;
- and readers interested in how systematic observation can contribute to problem solving outside formal research environments.

The investigation also illustrates the importance of preserving uncertainty. Throughout the record, unresolved questions are retained rather than replaced with unsupported conclusions, allowing future evidence to expand or revise the current working model.

1.9 Investigative Perspective

This document is neither a memoir nor a treatment manual.

It is an investigative record.

Where personal experiences are described, they are included because they constitute evidence relevant to the investigation rather than for narrative effect. Likewise, improvements are reported because they informed the evolving working model, not because they establish proof of a particular mechanism.

Accordingly, readers are encouraged to evaluate the investigation as they would any other longitudinal case record: by distinguishing direct observations from interpretations, considering the limitations under which the evidence was collected, and recognizing that the current working model represents the best explanation supported by the available evidence rather than a final or exhaustive account of the underlying mechanisms.

SECTION 2 — SUBJECT BACKGROUND

2.1 Subject Profile

The subject is an adult male with a lifelong interest in physical development, strength training, and athletic performance. Although consistently physically active throughout childhood and adulthood, his training history was characterized by repeated interruptions resulting from employment demands and later significant medical complications.

The investigation documented in this report arose from the convergence of these long-standing interests and the progressive emergence of persistent functional limitations that conventional training methods failed to resolve.

This report concerns a single individual and is therefore presented as a longitudinal self-case investigation rather than an attempt to establish generalizable clinical findings.

2.2 Early Athletic History

From early childhood, the subject demonstrated unusually high levels of physical capability despite maintaining a relatively lean physique. Athletic participation included running, jumping, recreational sports, and physically demanding daily activities. Strength was consistently described by others as disproportionate to body size.

Examples recalled by the subject include independently lifting and moving household furniture, mattresses, box springs, washing machines, and other heavy objects at ages when such tasks would ordinarily require assistance. These observations are presented not as quantitative measurements of strength but as representative examples of functional capacity during that period.

Despite possessing considerable natural strength and athletic ability, the subject consistently viewed his physical development as incomplete because of difficulty acquiring muscular body mass. Improving muscular size remained an enduring personal objective from childhood onward.

The only athletic limitation consistently perceived during this period involved cardiovascular endurance, although the subject acknowledges that this assessment reflected unusually high personal expectations rather than objective evidence of impaired physical performance.

Overall, the pre-investigation period was characterized by:

- high athletic capacity;
- unusually high relative strength;
- unrestricted participation in physical activities;
- normal confidence in movement;

- and the absence of the movement restrictions that would later become central to the investigation.

2.3 Lifelong Hard-Gainer History

The subject consistently identified as a "hard gainer," reporting persistent difficulty increasing body mass despite maintaining a strong appetite and prolonged efforts to gain muscular size.

This pattern was present throughout childhood, adolescence, and adulthood. According to the subject's recollection, no period of increased food intake resulted in sustained weight gain proportional to effort. Consequently, resistance training was pursued not merely as a recreational activity but as part of a long-standing effort to achieve desired muscular development.

Although later medical events substantially altered the direction of the investigation, the original motivation for strength training remained unchanged: to develop greater muscular size while preserving functional athleticism.

This lifelong context is important because the investigation did not originate from rehabilitation as a primary objective. Rather, rehabilitation emerged as a necessary prerequisite for returning to the original goal of aesthetic physical development.

2.4 Resistance Training History

The subject first acquired home resistance-training equipment at approximately sixteen years of age and began pursuing muscular development with greater intentionality during late adolescence.

Over subsequent decades, resistance training occurred in recurring periods of sustained participation interrupted by circumstances largely unrelated to motivation. These interruptions most commonly resulted from employment changes and other practical constraints on training continuity.

The longest uninterrupted period of consistent resistance training occurred between approximately 2013 and early 2014. During that period, the subject experienced what was initially believed to be a rotator cuff injury affecting the posterior shoulder region. Although the precise anatomical structure remains uncertain, this event represented one of the earliest significant musculoskeletal setbacks documented within the broader training history.

Despite repeated interruptions, the subject consistently returned to resistance training whenever circumstances permitted, reflecting a sustained commitment to long-term physical development rather than intermittent recreational participation.

2.5 Progressive Functional Decline

For much of early adulthood, the subject reported no significant movement restrictions beyond the longstanding difficulty gaining muscular size.

This pattern changed following entry into prolonged occupational driving.

Within months of beginning that occupation, the subject experienced the first of several progressively more significant physical problems affecting different regions of the body. Initially these problems appeared unrelated. However, over subsequent years they accumulated into an increasingly complex pattern involving the upper extremity, trunk, pelvis, and lower extremity.

At the time each individual problem emerged, no unifying explanation was apparent. Most were initially interpreted as isolated events requiring independent management.

Only much later, after systematic investigation had begun, did the possibility emerge that at least some of these seemingly separate problems might represent interconnected functional disturbances rather than unrelated injuries.

The chronology and evolution of these events are documented in detail in later sections of this report.

2.6 Circumstances Leading to the Investigation

The investigation described in this document did not begin immediately after the onset of physical symptoms.

For several years, the subject approached each emerging problem in a conventional manner, assuming that rest, recovery, and eventual return to resistance training would restore previous function. This expectation became increasingly difficult to maintain as additional symptoms accumulated and conventional approaches failed to produce complete recovery.

Following the diagnosis of deep vein thrombosis (DVT) and thoracic outlet syndrome (TOS) in 2019, resistance training was discontinued for approximately five years because the upper body remained painful, restricted, and susceptible to recurrent swelling during physical exertion.

During this period, the subject accepted physically demanding warehouse employment after leaving prolonged occupational driving. This work required prolonged standing, lifting, carrying, and repetitive lower-extremity loading during a period in which recovery from earlier lower-body dysfunction remained incomplete.

When the opportunity to return to the gym finally arose in November 2024, the subject intentionally resumed training only the lower body. This decision was not based on preference but on necessity. The upper body continued to exhibit pain, restriction, and exercise-induced swelling, making comprehensive resistance training impractical.

The expectation at that time was straightforward: restore lower-body strength through conventional training, then gradually resume normal physical development.

Instead, training repeatedly exposed persistent asymmetries, incomplete muscular participation, altered force transmission, and movement patterns that conventional strengthening alone failed to explain.

Those repeated observations marked the beginning of the formal investigation documented throughout the remainder of this case file.

SECTION 3 — METHODOLOGY

3.1 Investigative Approach

This investigation employed a longitudinal observational methodology built upon repeated self-observation, structured experimentation, iterative hypothesis development, and continuous refinement of working models over time.

Rather than following a predetermined experimental protocol, the investigation evolved organically in response to observations made during daily movement, resistance training, and deliberate functional testing. Each new finding informed subsequent experimentation, allowing the investigative process to adapt as understanding improved.

The investigation therefore emphasized discovery through repeated interaction with the problem rather than validation of an existing theory.

3.2 Initiation of the Investigation

The formal investigative phase began in **January 2025**, approximately two months after the subject resumed lower-body resistance training in November 2024.

The original expectation was that conventional strength training would restore normal lower-body function. Instead, progress plateaued despite continued effort. Increasing training volume or intensity without understanding the underlying limitation was considered unlikely to produce a different outcome.

This plateau marked the transition from conventional training to systematic investigation.

The emphasis shifted from asking:

"How can I become stronger?"

to asking:

"Why is normal strengthening not producing normal function?"

3.3 Investigative Philosophy

The investigation was guided by several foundational principles that remained consistent throughout the project.

These principles included:

- Do not assume; observe.

- Separate observations from explanations.
- Report findings before interpreting them.
- Compare observations against known or previously established function whenever possible.
- Retain uncertainty until sufficient evidence supports revision.
- Allow evidence—not preference—to determine the direction of the investigation.

Repeatedly throughout the investigation, observations were permitted to challenge existing explanations. Working models were regarded as temporary explanatory frameworks rather than conclusions.

This philosophy is summarized by the investigative cycle that gradually emerged through practice:

Observe → Compare → Experiment → Repeat → Retain what persists → Discard what does not → Refine the working model → Repeat

Figure 8. Investigative Cycle

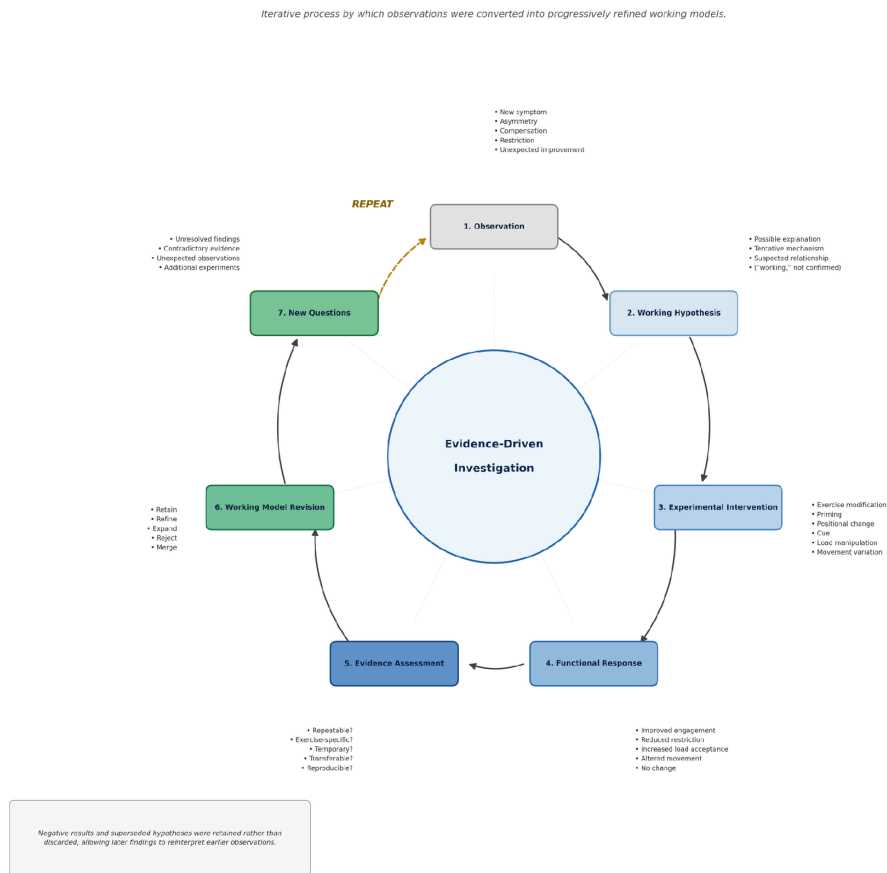


Figure 8. Investigative Cycle. Conceptual representation of the iterative process used throughout the investigation. Repeated observations generated working hypotheses that were tested through targeted exercise modifications and movement experiments. Functional responses were evaluated for repeatability and incorporated into progressively refined working models, which in turn generated new investigative questions. The cycle emphasizes that the investigation proceeded through continuous evidence-based refinement rather than through isolated discoveries.

Figure 8. Investigative Cycle

3.4 Experimental Process

Most experiments originated from a specific unanswered functional question rather than from curiosity alone.

Examples included:

- How can the hamstrings and gluteal musculature be consistently engaged during Romanian deadlifts?
- Why does the left side fail to accept load in the same manner as the right?
- What conditions improve or reduce movement restrictions?
- Why do some improvements persist while others disappear?

- Which regions participate before, during, and after specific interventions?

When a movement, cue, or intervention produced an unexpected result, additional testing was performed to determine whether the observation represented a transient event or a reproducible finding.

The investigation therefore progressed through successive cycles of observation, experimentation, and refinement rather than isolated experiments.

3.5 Experimental Interventions

Experimental interventions were intentionally simple and highly adaptable.

Common investigative strategies included:

- reducing external load to improve movement perception;
- modifying exercise technique;
- changing movement tempo;
- adjusting body position;
- introducing preparatory or "priming" exercises;
- stretching to explore movement restrictions;
- self-massage when appropriate;
- comparing left and right sides under similar conditions;
- and deliberately reproducing previously observed movement patterns.

No intervention was considered successful merely because it produced improvement. Rather, interventions were valued according to the information they revealed about the larger functional problem.

3.6 Decision Criteria

Throughout the investigation, observations were evaluated according to their explanatory value rather than their novelty.

Findings were generally retained when they demonstrated one or more of the following characteristics:

- reproducibility;
- logical consistency with previous observations;
- meaningful functional improvement;
- explanatory value;

- or integration into the developing working model.

Conversely, hypotheses were modified or abandoned when accumulating evidence demonstrated that they no longer adequately explained observed findings.

The duration required before accepting a finding varied according to the nature of the observation. Some findings were immediately compelling and subsequently confirmed through continued use. Others required repeated testing over days or weeks before they were incorporated into the working model.

No predetermined minimum number of repetitions was imposed. Confidence increased through continued successful reproduction rather than through adherence to a fixed testing schedule.

3.7 Use of Resistance Exercise as an Investigative Tool

Resistance exercises were not used solely for strength or hypertrophy.

Over time, they became investigative instruments.

Individual exercises evolved into environments within which specific functional questions could be examined under controlled loading conditions.

Exercises served multiple investigative roles, including:

- generating new observations;
- testing existing hypotheses;
- confirming previous discoveries;
- preparing specific movement systems ("priming");
- teaching coordinated movement ("integration");
- assessing persistence of prior improvements;
- and continuing physical development.

Consequently, exercises were evaluated according to their evidentiary contribution as well as their training value.

3.8 Load Selection Philosophy

One of the defining methodological characteristics of the investigation was the deliberate use of relatively light loads during exploratory phases.

Light loading frequently permitted movement quality, muscular participation, and force transmission to be perceived more clearly than heavier resistance.

The subject concluded that:

Light loads reveal organization; heavy loads reveal performance.

This principle guided numerous later experiments, particularly during periods focused on movement quality rather than maximal strength.

3.9 Interpretation of Pain and Restriction

Pain was not automatically interpreted as a signal to terminate investigation.

Instead, pain was treated as information requiring interpretation.

When pain occurred, movement was typically modified or temporarily discontinued while the response was observed. The objective was to determine:

- whether the pain represented tissue overload,
- mechanical compensation,
- protective behavior,
- or another reproducible phenomenon requiring further investigation.

This approach did not involve intentionally provoking injury. Rather, it reflected an effort to understand persistent functional limitations by carefully examining the circumstances under which symptoms appeared or resolved.

3.10 Role of Ingenuity

Because formal laboratory resources were unavailable, the investigation relied heavily upon practical ingenuity.

New experiments frequently emerged from previous observations rather than external protocols.

Exercises were modified, combined, or repurposed according to the questions being investigated.

Many of the investigation's most important discoveries resulted from adapting conventional movements to answer increasingly specific functional questions.

3.11 Professional Resources

This investigation was conducted independently.

No sports medicine laboratory, biomechanics laboratory, rehabilitation research facility, or dedicated movement-analysis equipment was available throughout the investigation.

Although medical care was sought when medically necessary, the functional investigation itself remained self-directed.

In retrospect, collaboration with professionals specializing in sports rehabilitation or movement analysis would likely have accelerated portions of the investigation and provided additional objective measurements.

Nevertheless, the absence of those resources contributed to the development of a highly observational, evidence-driven investigative process that forms the basis of this report.

3.12 Methodological Limitations

This document presents a longitudinal self-case investigation documenting the evolution of a persistent unilateral movement dysfunction over an extended period of observation. While the investigation generated a substantial body of internally consistent observations and progressively refined working models, several methodological limitations should be recognized when interpreting the findings presented throughout this document.

These include:

- this investigation represents a single-subject ($n = 1$) case study; the observations, interventions, and resulting working models describe one individual's presentation and should not be assumed to generalize to broader populations without independent investigation and replication;
- the absence of laboratory instrumentation and lack of continuous external measurement;
- reliance upon repeated self-observation during structured exercise, functional testing, and daily activity; although observations were documented contemporaneously whenever practical and repeatedly evaluated for consistency over time, self-observation inherently introduces potential observer bias and cannot provide the degree of objectivity associated with blinded or externally controlled investigations;
- inability to directly visualize anatomical structures during functional testing; several proposed mechanisms remain functional rather than anatomical explanations, and the precise anatomical structures responsible for many observations have not been confirmed through imaging, electrodiagnostic testing, cadaveric investigation, or comparable objective measures — accordingly, the Current Multi-Deficit Systems Model should be interpreted as the best explanation supported by the available evidence rather than as a confirmed anatomical diagnosis;
- an adaptive rather than predetermined experimental protocol; new observations frequently redirected subsequent experimentation, resulting in a methodology intentionally designed to preserve investigative responsiveness rather than strict experimental standardization, which necessarily limits direct comparison between some experimental periods; and
- the investigation remains active; several functional relationships identified within the current working model require additional study before they can be considered fully resolved. Rather than presenting the current model as definitive, this Case File

intentionally preserves both the strongest supported findings and the remaining unresolved questions so that future observations may refine, expand, or revise the present understanding without altering the historical investigative record.

Despite these limitations, the investigation benefited from extensive temporal continuity, repeated experimentation under comparable conditions, detailed contemporaneous documentation, systematic preservation of unsuccessful hypotheses alongside successful findings, explicit separation between confirmed observations, working interpretations, and unresolved questions, and continual refinement of the working model as new evidence accumulated. These characteristics provide a transparent evidentiary framework upon which future investigation may reasonably build.

SECTION 4 — CHRONOLOGY OF SIGNIFICANT EVENTS

4.1 Purpose

This chronology reconstructs the major events preceding and occurring during the investigation in their approximate historical sequence.

Its purpose is to establish temporal context for the evidence presented throughout the remainder of this case file. Whenever possible, dates are based upon contemporaneous records or events that can be reasonably anchored in time. Where exact dates cannot be established, the chronology reflects the best reconstruction supported by the available evidence.

The chronology documents **events**, not explanations. Interpretation of these events is reserved for the Case Files and the Evolution of the Working Model.

4.2 Chronology Confidence Statement

The chronology presented below combines documented dates with retrospective reconstruction.

Chronological entries are therefore assigned one of three confidence levels:

- **High Confidence** — Supported by documented dates or events that establish the timing with reasonable certainty.
- **Moderate Confidence** — Supported by consistent recollection anchored to known events but lacking exact documentation.
- **Approximate** — Estimated from the available evidence and subject to future refinement.

Refinement of dates in future versions shall not be interpreted as revision of the underlying observations.

4.3 Chronological Record

Approximate Date	Event	Confidence
1977	Birth of subject.	High
Age ~16	Acquired first weight bench and began structured resistance training at home.	Moderate
~1995	Developed a sustained personal interest in muscular development despite naturally lean physique.	Moderate
Late adolescence (18–19 years)	Began pursuing muscular size and strength with greater intentionality.	Moderate
2013 – Early 2014	Longest uninterrupted period of resistance training. Posterior shoulder injury (initially believed to involve the rotator cuff) occurred during this period.	High
September 2016	Experienced severe left scapular muscle spasm while engaged in prolonged occupational driving. Require emergency department evaluation. Persistent residual knot remained thereafter.	High
May 2017	Bilateral wrist pain developed, commonly interpreted as carpal tunnel syndrome. Progressive gripping difficulty affected occupational tasks.	Moderate
Winter 2018	Developed left gluteal and lower-back pain commonly described as sciatica. Walking, sitting, and prolonged positioning became increasingly painful.	Moderate
June 2019	Progressive pain developed involving the left pectoral region, anterior shoulder, and lateral axillary region. Upper-extremity function progressively deteriorated.	High
December 2019 (early)	Continued resistance training despite increasing symptoms. Final	High

Approximate Date	Event	Confidence
	upper-body training sessions preceded marked worsening of symptoms.	
December 2019 (approximately one to two weeks later)	Acute left upper-extremity swelling prompted emergency department evaluation. Deep vein thrombosis (DVT) and thoracic outlet syndrome (TOS) diagnosed.	High
December 2019	Resistance training discontinued.	High
September 2021	Entered physically demanding warehouse employment after leaving prolonged occupational driving. Occupational demo required prolonged standing, lifting, carrying, and repetitive lower-extremity loading during recovery.	High
December 2021	Completed prescribed anticoagulant therapy after determining that treatment would reduce future clot formation but not remove the existing clot. Continued daily low-dose aspirin thereafter.	Moderate
February 2022	Progressive upper-body symptoms stabilized sufficiently to permit gradual return to normal daily activity, although significant restrictions remained.	Moderate
November 2024 (mid-month)	Returned to gym after approximately five years. Training intentionally limited to the lower body because upper-body loading continued to provoke pain and swelling.	High
January 2025	Conventional lower-body strengthening reached an unexpected plateau. Formal investigation initiated.	High
2025–2026	Progressive experimental phase. Systematic observation, exercise modification, and working-model	High

Approximate Date	Event	Confidence
	development became increasingly structured.	
Spring 2026	Major proximal release events occurred, followed by progressive distal foot and ankle release sequence.	High
Spring–Summer 2026	Priming procedures, load-acceptance concepts, pelvic integration strategies, and exercise-specific investigative methods emerged.	High
July 2026	Recognition of the importance of knee/thigh stabilization and deliberate foot pressure during leg press.	High
August 2026	Horizontal leg press initiated the observation sequence leading to the multidirectional ankle-control discovery.	High
August 28, 2026	Integrated lower-body training session demonstrated the strongest left-chain organization observed to date and resulted in formulation of the current multi-deficit working model. This date serves as the evidentiary endpoint for Case File v1.0 .	High

4.4 Chronological Summary

Several characteristics become apparent when the chronology is viewed as a whole.

First, the major functional problems did not appear simultaneously. Instead, they emerged progressively over several years, initially affecting anatomically distinct regions that appeared unrelated.

Second, the transition from conventional resistance training to formal investigation did not occur immediately following the onset of symptoms. Rather, it occurred only after conventional strengthening repeatedly failed to restore expected function.

Finally, the investigation itself evolved through identifiable phases. Early efforts focused primarily on restoring individual muscles and regions. Later phases increasingly emphasized coordinated movement, load acceptance, force transmission, and systems-level organization, ultimately culminating in the current multi-deficit working model.

This chronology provides the temporal framework within which the detailed evidence presented in the subsequent Case Files should be interpreted.

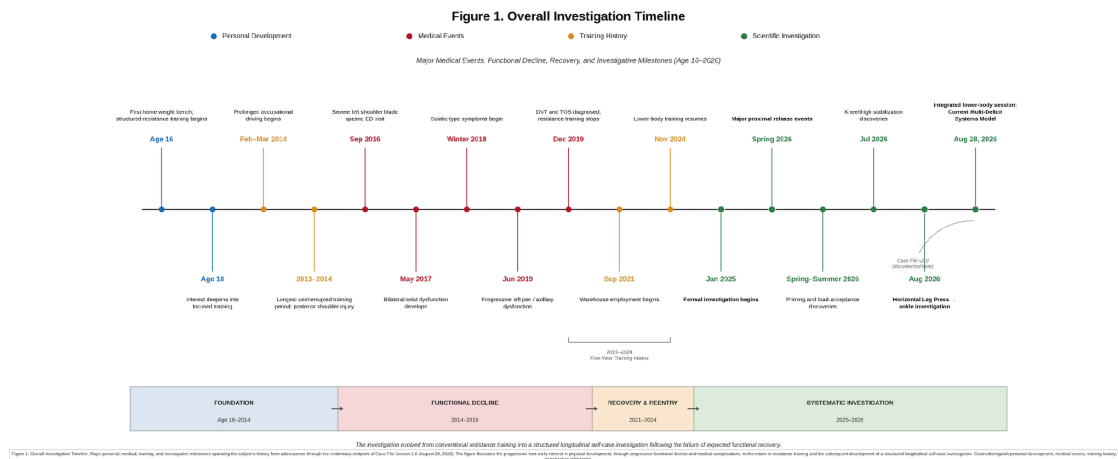


Figure 1. Overall Investigation Timeline

SECTION 5 — INVESTIGATIVE EVIDENCE RECORDS

5.1 Purpose

This section constitutes the primary evidentiary foundation of the investigation.

The records presented herein document the observations, experiments, exercise findings, and functional changes that materially influenced the evolution of the investigation. Collectively, they represent the body of evidence from which all subsequent interpretations and working models were developed.

No attempt is made within this section to retrospectively reorganize earlier findings according to later conclusions. Instead, each record is preserved within its original investigative context so that the progression of understanding remains transparent and auditable.

The objective of this section is to present the evidence itself. Interpretation and synthesis of that evidence are reserved for subsequent analytical sections of this report.

5.2 Organization of the Evidence

The investigative record is divided into two complementary evidence collections.

Case Files (CF)

Case Files document discrete observations, events, discoveries, or investigative milestones that materially influenced the direction of the investigation.

Each Case File follows a standardized structure including:

- chronology;
- event;
- context;
- direct observations;
- functional consequences;
- interventions;
- immediate outcomes;
- delayed outcomes;
- interpretation at the time;
- current interpretation;
- remaining uncertainties;
- related records;
- and confidence assessment.

Case Files primarily document **what happened** during the investigation.

Exercise Evidence Records (ER)

Exercise Evidence Records document the investigative role of individual exercises.

Unlike conventional training logs, these records treat exercises as experimental environments rather than merely strength or hypertrophy movements.

Each Exercise Evidence Record documents:

- why the exercise was introduced;
- how its technique evolved;
- discoveries produced;
- hypotheses generated or tested;
- current investigative role;
- evidentiary importance;
- and present disposition within the investigation.

Exercise Evidence Records primarily document **how evidence was generated**.

5.3 Relationship Between the Evidence Collections

Although Case Files and Exercise Evidence Records are presented separately, they should be interpreted as complementary components of the same investigative record.

Case Files frequently reference exercises responsible for important observations.

Likewise, Exercise Evidence Records frequently reference Case Files documenting the discoveries those exercises produced or confirmed.

Neither collection should therefore be interpreted independently of the other.

Together they provide both:

- a chronological record of the investigation; and
- a methodological record of how the evidence was produced.

5.4 Interpretation of Evidence

Readers are encouraged to distinguish carefully between:

- observed events;
- experimental interventions;
- functional consequences;
- and explanatory interpretations.

Throughout this section:

- observations are preserved regardless of later interpretation;
- interpretations reflect the best explanation supported by the evidence available at that stage of the investigation;
- unresolved questions remain explicitly identified;
- and confidence assessments indicate the degree of certainty associated with individual findings.

The purpose of preserving this distinction is to allow future evidence to refine or expand the investigation without altering the historical record.

5.5 Evidence Classification

Unless otherwise indicated, every record within this section is interpreted according to the Evidence Classification System established in the Front Matter.

Observations are therefore distinguished as:

- Confirmed Medical Diagnosis
- Direct Observation
- Working Interpretation / Working Hypothesis
- Unknown

Exercise Evidence Records include an additional editorial classification identifying each exercise according to its investigative contribution.

These classifications include:

- Primary Discovery Exercise
- Major Supporting Exercise
- Confirmatory Exercise
- Integration Exercise
- Priming Exercise
- Development Exercise
- Assessment Exercise

Because many exercises fulfilled multiple investigative roles over time, individual Exercise Evidence Records may contain more than one classification.

Figure 9. Exercise Classification Map

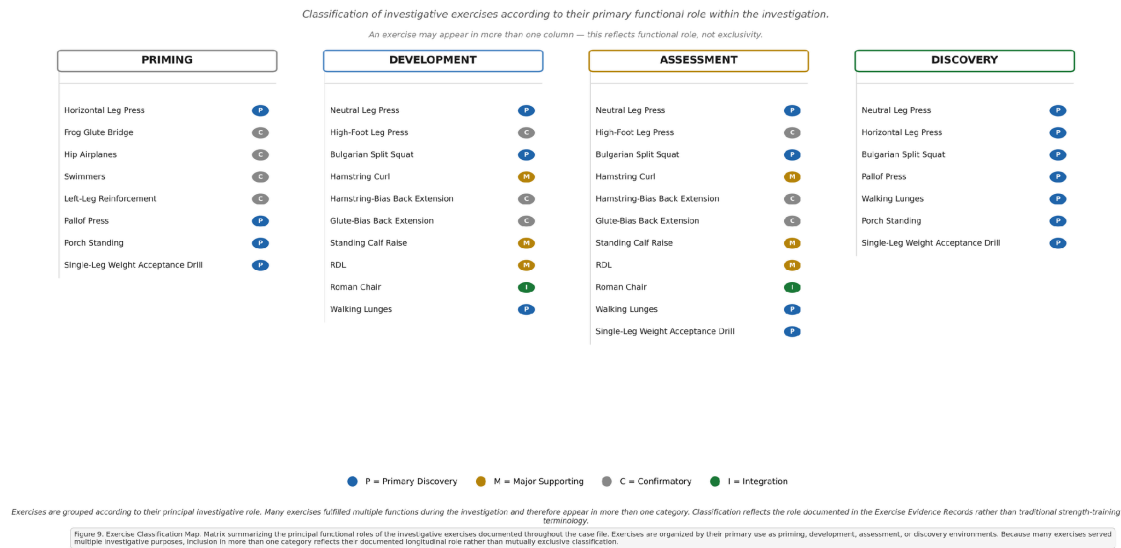


Figure 9. Exercise Classification Map

5.6 Evidence Sequence

The primary evidence is presented in the following order:

Part I — Case Files

CF-01 — Pre-existing Left Adductor Injury

1. Date

Approximately 2025. Exact date currently unestablished. On April 26, 2026, injury was described as occurring approximately one year earlier.

2. Event

Left adductor was injured while using an adductor machine.

3. Context

Resistance-training injury during adductor-machine use.

4. Direct observations

- Acute left-adductor injury.
- Pain persisted for months.
- Tightness/weakness remained after the acute period.
- Knee-to-chest movement on the left became restricted.

- Restriction affected movements requiring deep hip flexion, including presses/lunges.

5. Functional consequences

- Reduced left hip-flexion availability.
- Persistent restriction affecting lower-body training.
- Became one component of the larger left-chain problem.

6. Interventions performed

Multiple later mobility, priming, strengthening, and reintegration interventions; no single intervention established as the treatment for the original injury.

7. Immediate outcomes

Not sufficiently documented.

8. Delayed outcomes

Persistent tightness/restriction remained long after acute pain.

9. Medical evaluation/diagnosis

No confirmed medical diagnosis for the adductor injury established in the current case record.

10. Interpretation at the time

Adductor pull/injury.

11. Current interpretation

The prior adductor injury remains relevant to the left-chain history, but the entire later dysfunction is **not attributed solely to the adductor injury**.

12. Remaining uncertainties

- Exact injured structure.
- Injury severity.
- Exact date.
- Whether dysfunction elsewhere preceded the injury.
- Degree to which the injury initiated versus amplified later chain dysfunction.

13. Related events

CF-02, CF-03, CF-10, CF-16.

14. Confidence

High for occurrence and persistent symptoms; **Low-Medium** for anatomical mechanism.

CF-02 — Development/Recognition of Global Left-Chain Dysfunction

1. Date

Gradual; exact onset not established.

2. Event

Recognition that the problem extended beyond the original adductor region and involved the left lower-body chain as a coordinated system.

3. Context

Walking, sitting, hinging, unilateral loading, bilateral loading, and deliberate movement testing.

4. Direct observations

- Left side did not accept load like the right.
- Left leg did not naturally stabilize/catch the body during walking like the right.
- Push-off was deficient.
- Downward pressure through the left side could not be produced normally.
- Left thigh did not brace normally.
- Pelvis tended to escape/drop away from the left side under some loading conditions.
- Knee, shin/lower leg and other regions compensated.
- Glute engagement was impaired.
- Hip hinging was restricted.
- Foot/ankle behavior differed substantially from the right.

5. Functional consequences

- Abnormal gait/load acceptance.
- Reduced unilateral stability.
- Altered force transmission.
- Restricted hinge.
- Difficulty maintaining integrated left-side engagement during resistance exercise.

6. Interventions performed

Subsequent investigation involved foot, ankle, knee, thigh, hip, glute, pelvis and trunk rather than treating one muscle as the entire problem.

7. Immediate outcomes

Not applicable as a single event.

8. Delayed outcomes

Progressively more components of the chain became consciously accessible.

9. Medical evaluation/diagnosis

No medical diagnosis of a unified “left-chain dysfunction” is established. **“Left chain offline” is the user's functional description/model.**

10. Interpretation at the time

Individual restrictions and weaknesses were initially being discovered separately.

11. Current interpretation

Several distinct **sub-issues appear to converge on one larger functional problem: the left chain being offline/incompletely incorporated.**

12. Remaining uncertainties

- Original initiating event(s).
- Relative contribution of each sub-issue.
- Whether additional unidentified restrictions remain.

13. Related events

All subsequent entries.

14. Confidence

High for functional asymmetry; **Medium** for current systems interpretation.

Figure 6. Left-Chain Functional Organization

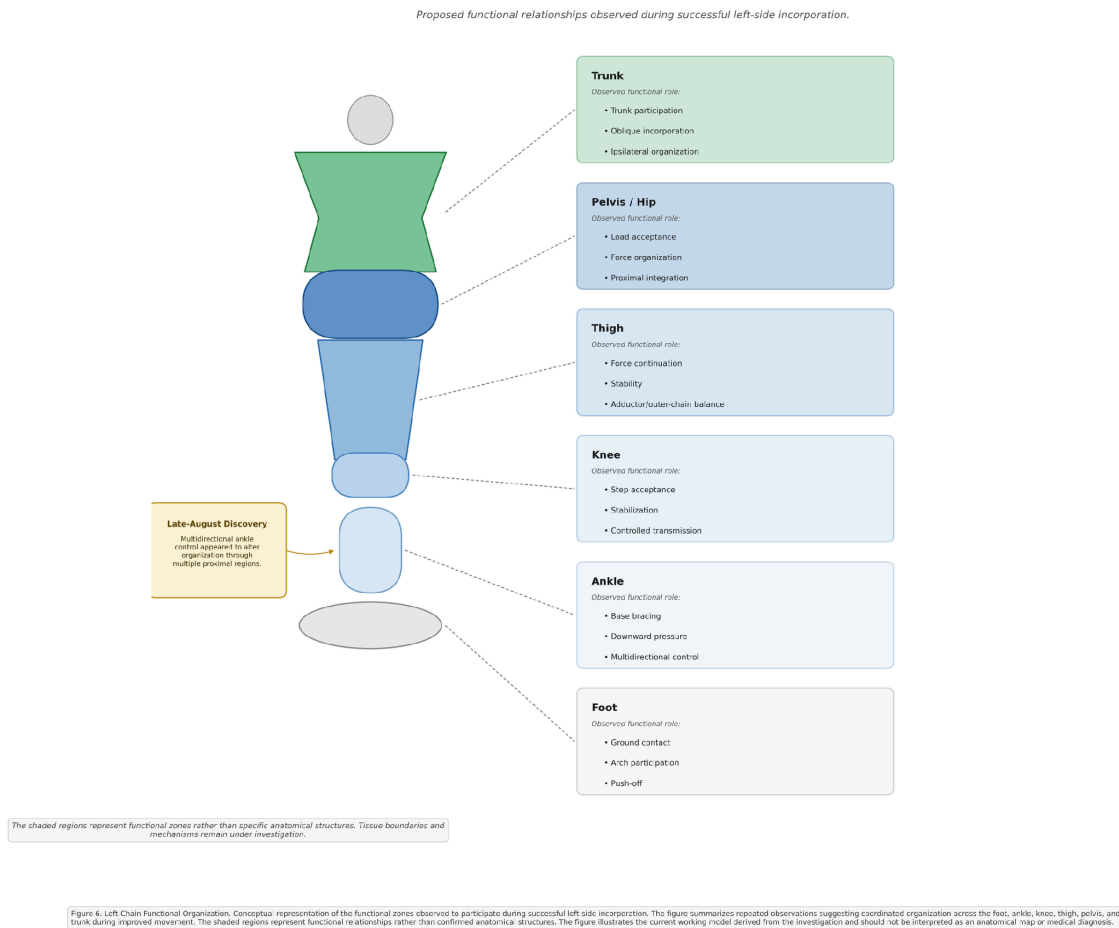


Figure 6. Left-Chain Functional Organization

CF-03 — Two Centers of Hip/Glute Restriction Identified

1. Date

Identified during 2026 investigation; exact first-identification date requires chronology refinement.

2. Event

Two recurring centers of restriction were distinguished.

3. Context

Hinging, stretching, sitting, massage, movement testing and lower-body training.

4. Direct observations

- One restriction centered around the lateral hip/glute region.
- A second centered near the sit-bone/proximal hamstring-lower-glute region.

- Both affected movement availability.
- Hip hinging remained restricted despite improvements elsewhere.

5. Functional consequences

- Restricted full hip hinging.
- Altered hip/pelvic mechanics.
- Difficulty fully accessing posterior-chain movement.

6. Interventions performed

- Massage.
- Heat/warm compress.
- Stretching.
- Hip airplanes.
- Controlled movement.
- Hinging experiments.
- Subsequent loading/priming work.

7. Immediate outcomes

Variable temporary releases and improved movement.

8. Delayed outcomes

Substantial improvement in hip/glute access, although restriction has not been established as completely resolved.

9. Medical evaluation/diagnosis

No confirmed diagnosis identifying the specific tissue responsible for either center.

10. Interpretation at the time

Tight/restricted tissue interfering with the hip/glute chain.

11. Current interpretation

These remain a **separate proximal branch** of the overall problem rather than an explanation for every distal deficit.

12. Remaining uncertainties

- Exact tissue(s).
- Whether one center influences the other.
- Relationship to original adductor injury.

13. Related events

CF-04, CF-05, CF-07, CF-17.

14. Confidence

High that two repeatable restriction centers are perceived; **Low-Medium** for anatomical identification.

CF-04 — Sitting-Induced Neurological Symptoms

1. Date

Documented during 2026; exact onset unknown.

2. Event

Prolonged sitting directly on one problematic glute region produced progressive sensory symptoms.

3. Context

Most clearly observed while sitting on a toilet seat.

4. Direct observations

- After approximately 10–15 minutes, the contact area became numb.
- Numbness/tingling subsequently extended into the lower leg, ankle and foot.
- With continued sitting, the hamstring could become numb.

5. Functional consequences

Prolonged sitting on that region is poorly tolerated.

6. Interventions performed

No definitive intervention established.

7. Immediate outcomes

Sensory symptoms occur with sustained compression/contact.

8. Delayed outcomes

Not sufficiently documented.

9. Medical evaluation/diagnosis

No confirmed diagnosis recorded.

10. Interpretation at the time

Initially discussed in connection with the problematic glute region.

11. Current interpretation

Mechanism remains **unknown**. Radiating numbness/tingling is maintained separately in the case file from muscular tightness because neurological involvement has not been excluded.

12. Remaining uncertainties

- Nerve(s) involved, if any.
- Exact compression site.
- Relationship to either restriction center.
- Relationship to broader left-chain dysfunction.

13. Related events

CF-03.

14. Confidence

High for reported symptoms; **Low** for mechanism.

CF-05 — Early Proximal Release Event / Glute Access

1. Date

April 25, 2026 or shortly before.

2. Event

Crossed-ankle seated leg-extension simulation produced painless popping/release around the hip/glute.

3. Context

Deliberate movement experimentation.

4. Direct observations

- Painless pop(s).
- Immediate change in left glute function.
- Left glute could subsequently be fully squeezed.
- Outer glute/hip resistance remained.

5. Functional consequences

Previously inaccessible glute contraction became available.

6. Interventions performed

Crossed-ankle seated leg-extension-type movement.

7. Immediate outcomes

Marked increase in voluntary glute contraction.

8. Delayed outcomes

Improvement persisted sufficiently to become part of subsequent reintegration work.

9. Medical evaluation/diagnosis

None established.

10. Interpretation at the time

Significant release within the proximal chain.

11. Current interpretation

A meaningful functional release event, but exact structure/mechanism remains unknown.

12. Remaining uncertainties

What physically produced the pop and why function changed afterward.

13. Related events

CF-06, CF-07.

14. Confidence

High for event/outcome; **Low** for anatomical explanation.

CF-06 — Distal Foot/Ankle Release Sequence

1. Date

Spring 2026; after CF-05.

2. Event

A sequence of ankle/foot pops occurred during continued experimentation.

3. Context

- Porch movement/stretching.
- Standing stiff-leg stretching.
- Continued proximal priming.
- Foot/ankle manipulation.

4. Direct observations

- An ankle pop occurred during a porch stiff-leg stretch.
- Additional foot/ankle pops followed over time.
- Some later pops were larger.
- Relief compounded after releases.
- Prior to this period, foot cramping occurred in the arch/pad/toe region, including toe-crossing behavior.
- Arch contact/load acceptance subsequently improved.

5. Functional consequences

- Better relationship with the ground.
- Reduced distal restriction.

- Improved foot participation in the chain.

6. Interventions performed

- Standing movement.
- Stretching.
- Foot/ankle movement.
- Continued proximal priming.

7. Immediate outcomes

Relief after individual pops/releases.

8. Delayed outcomes

Improved arch acceptance and later improvements in ankle/foot incorporation.

9. Medical evaluation/diagnosis

None established.

10. Interpretation at the time

Distal release appeared connected to improvements produced by proximal priming.

11. Current interpretation

Foot structural tightness remains one sub-issue. The later-discovered ankle muscular-control deficit is considered **related but distinct**.

12. Remaining uncertainties

- Structures producing pops.
- Whether releases were joint, soft tissue, or both.
- Exact relationship between proximal priming and distal release.

13. Related events

CF-05, CF-09, CF-18.

14. Confidence

High for observed releases; **Low** for mechanism.

CF-07 — Priming Procedure Emerges

1. Date

Spring–summer 2026.

2. Event

A repeatable collection of preparatory interventions developed.

3. Context

Repeated self-testing before walking/training and during periods of restriction.

4. Direct observations

Movement quality often improved after one or more of:

- massage of tight/sore areas,
- stretching,
- hip airplanes,
- warm compress,
- light controlled movement,
- treadmill walking,
- standing porch work,
- Pallof press stance,
- bodyweight single-leg RDL,
- horizontal leg press,
- standing calf raises.

5. Functional consequences

Improved access to movements and tissues before loaded work.

6. Interventions performed

As listed above.

7. Immediate outcomes

Variable improvements in movement, balance, engagement and load acceptance.

8. Delayed outcomes

Repeated priming became part of the ongoing reincorporation process rather than a discrete temporary “phase.”

9. Medical evaluation/diagnosis

Not applicable.

10. Interpretation at the time

Preparation helped unlock/recruit portions of the chain.

11. Current interpretation

Priming provides conditions under which deficient components can participate; exact mechanisms vary and remain partly unknown.

12. Remaining uncertainties

Which interventions are necessary versus merely helpful as automaticity increases.

13. Related events

CF-08 through CF-18.

14. Confidence

High for repeatable subjective functional effects.

CF-08 — Pallof Press and Hip-Airplane Findings

1. Date

Spring 2026.

2. Event

Pallof press and hip airplanes produced notable improvements in left-side integration.

3. Context

Corrective experimentation.

4. Direct observations

- Pallof press produced one of the strongest immediate improvements in single-leg balance.
- Counterweight/stance produced engagement up the leg.
- Hip airplanes produced meaningful hip/pelvic effects.
- A “Roman twist” cue during airplane work improved pelvic draw-in.

5. Functional consequences

- Improved balance.
- Improved pelvic/trunk participation.
- Better left-side organization.

6. Interventions performed

Pallof press and hip airplanes.

7. Immediate outcomes

Improved single-leg control.

8. Delayed outcomes

Pallof press retained as a trunk/pelvis integration exercise.

9. Medical evaluation/diagnosis

None.

10–11. Interpretation at the time/current interpretation

Useful integration tools rather than simple isolated “core” exercises.

12. Remaining uncertainties

Exact mechanism responsible for disproportionate response.

13. Related events

CF-09, CF-11, CF-18.

14. Confidence

High for functional response.

CF-09 — Arch Begins to “Own the Ground”

1. Date

Spring–summer 2026.

2. Event

Foot-ground relationship changed following repeated release and integration work.

3. Context

Standing, walking and loaded exercise.

4. Direct observations

- Arch increasingly felt as though it “owned the ground.”
- Left foot began attempting more normal push-off.
- Ground contact became more active rather than passive.

5. Functional consequences

- Improved walking mechanics.
- Better distal contribution to load acceptance.

6. Interventions performed

Foot/ankle work plus proximal priming.

7. Immediate outcomes

Improved ground interaction.

8. Delayed outcomes

Set groundwork for later recognition of ankle muscular control.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

Foot/arch becoming reincorporated.

11. Current interpretation

Arch improvement was important but did **not** completely resolve the base because multidirectional ankle control remained deficient.

12. Remaining uncertainties

Relationship between remaining foot tightness and ankle-control deficit.

13. Related events

CF-06, CF-11, CF-18.

14. Confidence

High for observed functional change.

CF-10 — Adductor/Hip Release and Transition Toward Strengthening**1. Date**

May 24, 2026.

2. Event

Leg press could be performed with zero descent tightness associated with the prior adductor/hip restriction.

3. Context

Lower-body training.

4. Direct observations

Previously present descent restriction was absent.

5. Functional consequences

Previously unavailable movement could now be loaded more normally.

6. Interventions performed

Prior release/priming/reintegration work.

7. Immediate outcomes

Improved descent.

8. Delayed outcomes

Attention began shifting from obtaining release toward strengthening newly available musculature.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

Major release milestone.

11. Current interpretation

Meaningful improvement, but not proof that all proximal restrictions were resolved.

12. Remaining uncertainties

Durability and precise anatomical mechanism.

13. Related events

CF-01, CF-03.

14. Confidence

High for functional change.

CF-11 — Knee/Thigh Catch and Push-Off Deficit

1. Date

Clearly documented July 7, 2026.

2. Event

Walking analysis revealed a major right-left difference.

3. Context

Walking and leg-press experimentation.

4. Direct observations

- Right side naturally caught/stabilized a step using knee/thigh/hamstring.
- Left side did not do so comparably.
- Left side historically lacked normal push-off.
- On leg press, pressing **into** the pad while simultaneously pressing/sliding the foot **down inside the shoe** increased anterior/medial-chain engagement.
- Engagement could be maintained more effectively through the eccentric.

5. Functional consequences

- Poor step acceptance/stability.
- Compensation elsewhere in left leg.

6. Interventions performed

Deliberate press-in/press-down cue.

7. Immediate outcomes

Improved engagement through knee/inner thigh/anterior chain.

8. Delayed outcomes

Contributed to increasingly complete load acceptance.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

Tibial/knee/femoral control and push-off were important to the chain.

11. Current interpretation

Still valid, but subsequent ankle discovery provides additional information about **why the distal base itself could not brace normally**.

12. Remaining uncertainties

Exact contribution of ankle control versus knee/thigh deficits.

13. Related events

CF-09, CF-12, CF-18.

14. Confidence

High for observed asymmetry and cue response.

CF-12 — Single-Leg Weight-Acceptance Drill

1. Date

Summer 2026.

2. Event

A unilateral bodyweight movement resembling a one-leg sissy squat was discovered experimentally.

3. Context

Non-lower training days / corrective work.

4. Direct observations

- Load perceived from below knee through ankle/foot.
- Repeated use improved subjective left-side load acceptance.

5. Functional consequences

Better ability to accept bodyweight through distal chain.

6. Interventions performed

Light unilateral weight-acceptance drill.

7. Immediate outcomes

Improved awareness/control.

8. Delayed outcomes

Retained as a useful corrective movement.

9. Medical evaluation/diagnosis

None.

10–11. Interpretation

A tool for teaching the left side to accept load rather than avoid it.

12. Remaining uncertainties

Exact contribution relative to simultaneous interventions.

13. Related events

CF-11, CF-14, CF-18.

14. Confidence
Medium-High.

CF-13 — Frog Bridge / Pelvic Integration Discovery

1. Date

Summer 2026.

2. Event

Frog glute bridges added to warm-up.

3. Context

Soles together, knees outward.

4. Direct observations

- Strong engagement through glute region into leg.
- Front pelvic/top hip-flexor area also perceived.
- Region felt increasingly integrated “as one unit.”

5. Functional consequences

Improved pre-training pelvic/hip engagement.

6. Intervention

Frog bridges.

7–8. Outcomes

Useful enough to retain as warm-up/integration work.

9. Medical evaluation

None.

10–11. Interpretation

Pelvic/hip reintegration tool.

12. Remaining uncertainties

Exact structures responsible.

13. Related events

CF-14, CF-15.

14. Confidence

High for subjective response.

CF-14 — Supermans / Swimmers Before Hamstring Curl

1. Date

Summer 2026.

2. Event

Prone leg-only Supermans and slow scissoring “Swimmers” used before lying hamstring curls.

3. Context

Hamstring preparation.

4. Direct observations

- Better posterior pelvic/hamstring engagement.
- Pelvic region felt engaged across front/back/side relationships.
- Subsequent curls improved.

5. Functional consequences

Better access to desired hamstring recruitment.

6. Interventions

- Leg-only Supermans.
- Slow paused Swimmers.
- Peak-hold hamstring warm-up.

7–8. Outcomes

Improved subsequent curl engagement.

9. Medical evaluation

None.

10–11. Interpretation

Useful neuromuscular primer.

12. Remaining uncertainties

Mechanism and necessity once automaticity improves.

13. Related events

CF-17.

14. Confidence

High for observed effect.

CF-15 — Horizontal vs. Linear Leg-Press Priming

1. Date

August 2026; horizontal-press breakthrough immediately precedes later ankle discovery.

2. Event

Different leg-press machines were found to provide different priming effects.

3. Context

Very light, controlled corrective loading.

4. Direct observations

Horizontal leg press

- Extremely light machine resistance allowed approximately 10–15 lb experimental loading.
- Pulses/half-reps could be performed while preserving trunk engagement.
- Strong proximal response.
- Perceived sequence included hip/trunk → outer quad/outer knee → main quad.
- Particularly useful for glute, pelvic/hip-flexor region and trunk of thigh.

Linear leg press

- Particularly useful for knee/lower-leg priming.

5. Functional consequences

Allowed selective preparation depending upon where the chain required assistance.

6. Interventions performed

Very light deliberate pressing and partial/pulse work.

7. Immediate outcomes

Improved chain access.

8. Delayed outcomes

- Horizontal press retained as a proximal primer.
- Possible hypertrophy role remains under investigation.
- Horizontal press directly preceded the ankle/foot discovery sequence.

9. Medical evaluation

None.

10. Interpretation at the time

Horizontal = proximal primer; linear = distal primer.

11. Current interpretation

Still retained. Horizontal pressing also appears to have created conditions that helped expose the later ankle-control relationship.

12. Remaining uncertainties

Whether horizontal press should also become a main hypertrophy movement.

13. Related events

CF-16, CF-18.

14. Confidence

High for differing perceived effects.

CF-16 — Bulgarian Split Squat Integration / Adductor Pinch

1. Date

August 2026, including August 28.

2. Event

Bulgarian split-squat mechanics improved substantially.

3. Context

Following successful priming.

4. Direct observations

- First notably successful session produced whole-leg-through-glute engagement.
- At times no adductor pinch occurred.
- When pinch appeared, maintaining/re-establishing outer-leg engagement could make it recede.
- Excessive load interfered with engagement, leading user to reduce weight.
- **August 28:** user could perceive driving force **from/through heel to/through hip, front and back.**

5. Functional consequences

- Improved unilateral force transmission.
- Better ability to preserve integrated left-side loading.

6. Interventions performed

- Priming.
- Deliberate outer-leg engagement.
- Load reduction when necessary.
- August 28 ankle/foot strategy.

7. Immediate outcomes

Improved integrated movement.

8. Delayed outcomes

Ongoing.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

Engagement prevented the adductor from becoming the dominant/problematic structure.

11. Current interpretation

Bulgarian split squat has become an important test of **ground-to-hip force transmission**.

12. Remaining uncertainties

- Mechanism of transient adductor pinch.
- Degree to which ankle control will make the improved pattern automatic.

13. Related events

CF-01, CF-15, CF-18, CF-19.

14. Confidence

High for observations; **Medium** for mechanism.

CF-17 — Proximal Hamstring “Strings” Identified Under Load

1. Date

August 28, 2026.

2. Event

Specific proximal hamstring region could be deliberately targeted.

3. Context

Hamstring curls and glute-biased back extensions.

4. Direct observations

- **30 lb hamstring curl** described as the “sweet spot” for the “strings in hams directly underneath butt-bone central.”
- A heavier 40-lb curl was possible, but 30 lb provided the desired targeting.
- Back extensions subsequently targeted the same region strongly; area described as “lit up & on fire.”

5. Functional consequences

Previously problematic proximal region could be loaded deliberately through more than one exercise.

6. Interventions performed

- Hamstring curl.
- Glute-biased back extension.

7. Immediate outcomes

Strong localized muscular engagement.

8. Delayed outcomes

Not yet established.

9. Medical evaluation/diagnosis

No anatomical diagnosis established for the perceived “strings.”

10–11. Interpretation at the time/current interpretation

Specific proximal hamstring/lower-glute-adjacent region appears increasingly accessible to deliberate loading.

12. Remaining uncertainties

- Exact tissue.
- Relationship to second restriction center.

13. Related events

CF-03, CF-14.

14. Confidence

High for repeatable sensation; **Low-Medium** for anatomical identification.

CF-18 — Multidirectional Ankle-Control Discovery

1. Date

Discovery sequence: approximately late August 2026.

Initial trigger: horizontal-leg-press priming **two sessions before August 28**.

Walking observation: several days afterward.

Home mapping/testing: same general interval.

Loaded application: **August 28, 2026**.

2. Event

Discovery that muscular exertion/control around the ankle in different directions changes perceived engagement higher in the ipsilateral chain.

3. Context

1. Horizontal leg-press priming produced the initial clue.
2. A subsequent walk produced another observation.
3. User developed a hypothesis.
4. Returned home and deliberately mapped/tested it.
5. Compared left behavior against right.
6. Applied the discovered strategy during August 28 training.

4. Direct observations

- Muscular actions used to move/control the foot at the ankle **up, down, left and right** correspond with different perceived engagement through the glute/trunk.
- Right ankle can provide much fuller “360°” control/bracing.
- Left does not yet perform this automatically.
- Intentional left ankle control changed whole-chain engagement.
- Application during August 28 training worked.
- Standing calf raises produced whole-leg trembling on each rep/set while the entire chain through ankle and arch was perceived as engaging.

5. Functional consequences

- Improved base bracing.
- Improved ability to incorporate ankle/arch into whole-leg loading.
- Provides a possible explanation for several longstanding functional deficits.

6. Interventions performed

- Deliberate multidirectional ankle muscular exertion/control.
- Right-left comparison.
- Home testing.
- Application during loaded exercise.

7. Immediate outcomes

- Improved perceived chain recruitment.
- Successful incorporation during August 28 session.

8. Delayed outcomes

Not yet established. Automaticity is the next functional issue under observation.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

Initial clues were not understood.

11. Current interpretation

Ankle muscular-control deficit represents **one distinct sub-issue** contributing to the larger offline-chain problem.

It does **not** explain:

- the two proximal restriction centers,
- all hip-hinge restriction,
- all foot tightness,
- or the entire original dysfunction.

It does potentially explain important portions of:

- inability to brace at the base,
- inability to “grab” the ground during walking,
- inability to produce normal downward pressure,
- and incomplete distal-to-proximal force transmission.

12. Remaining uncertainties

- Exact musculature involved.
- Exact relationship between individual ankle directions and specific proximal recruitment.
- Whether perceived mappings generalize across exercises.
- How much is ankle-specific versus coupled foot/tibial/femoral movement.
- Time required for automaticity.

13. Related events

CF-06, CF-09, CF-11, CF-15, CF-19.

14. Confidence

High that intentional ankle-control changes altered perceived chain engagement;

Medium-Low for exact biomechanical explanation.

CF-19 — Retrospective Reinterpretation of “Cannot Apply Downward Pressure”

1. Date

Original observation predates August 28, 2026; exact first date currently unresolved.

Reinterpretation occurred August 28.

2. Event

A longstanding inability to apply downward pressure through the left side was reinterpreted following CF-18.

3. Context

This inability had been one of the major unresolved functional problems driving repeated experimentation.

4. Direct observations

- User had long recognized inability to apply normal downward pressure.
- Left side could not brace at the base like right.
- Ground could not be “grabbed” normally during walking.
- This unresolved problem motivated repeated porch experiments, ankle flexion/manipulation and other trials.
- User did **not** understand the mechanism at the time but deliberately continued investigating it.

5. Functional consequences

- Deficient ground interaction.
- Poor push-off.
- Poor base bracing.
- Incomplete load acceptance/transfer.

6. Interventions performed

- Porch experimentation.
- Repeated ankle flexion/manipulation.
- Walking observation.
- Foot/ankle work.
- Leg-press experimentation.
- Multiple chain-reintegration exercises.

7. Immediate outcomes

Individual experiments produced partial clues rather than an immediate explanation.

8. Delayed outcomes

CF-18 eventually supplied a working explanation for an important portion of the deficit.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

A clearly perceived functional deficit with unknown cause.

11. Current interpretation

The ankle's deficient muscular bracing/control appears to explain **why the base could not organize sufficiently for normal downward pressure**, while other independent sub-issues simultaneously affected other portions of the chain.

12. Remaining uncertainties

- How much of original downward-pressure deficit was specifically ankle-derived.
- Whether remaining foot restriction also contributes.
- Whether other distal deficits remain unidentified.

13. Related events

CF-02, CF-06, CF-09, CF-11, CF-18.

14. Confidence

High that downward-pressure deficit was longstanding and drove investigation; **Medium** for current causal interpretation.

CF-20 — August 28 Integrated Lower-Body Session

1. Date

August 28, 2026.

2. Event

Lower-body session produced several simultaneous indications of improved left-chain incorporation.

3. Context

Loaded Lower B training after recent ankle discovery and application.

4. Direct observations

Neutral Leg Press

- Left trunk fully engaged.
- Left side accepted load throughout full ROM.
- Reps intentionally very slow/deliberate.
- Load acceptance was perceived extending **through pelvis to oblique**.

Bulgarian Split Squat

- Driving force perceived **from/through heel to/through hip, front and back**.

Hamstring Curl

- 30 lb identified as sweet spot for proximal “strings” immediately below central sit-bone area.

Back Extension

- Same proximal hamstring region strongly targeted.

Standing Calf Raise

- Whole leg trembled on every rep/set.
- Trembling occurred while entire chain through **ankle and arch** was perceived as engaging.

Session-Level Observation

- User estimated left trunk accepted approximately **95–98% of its portion of the load throughout the session.**
- A brief adductor pinch occurred.
- Surrounding musculature then re-engaged and the pinch subsided.

Important evidence qualifier: 95–98% is a **subjective proprioceptive estimate**, not an instrument-measured load distribution.

5. Functional consequences

- Load acceptance persisted across multiple exercises.
- Distal, proximal and trunk participation were simultaneously accessible.
- Force pathway could be perceived during unilateral loading.
- Brief loss of desired organization may have been followed by spontaneous/rapid reorganization.

6. Interventions performed

- Previously developed priming.
- Newly discovered ankle-control strategy.
- Deliberate slow loading.
- Exercise-specific engagement strategies.

7. Immediate outcomes

- Strongest reported session-wide left-side load acceptance to date in the available record.
- Successful application of ankle discovery under meaningful load.

8. Delayed outcomes

Unknown as of August 28.

9. Medical evaluation/diagnosis

None.

10. Interpretation at the time

Multiple previously isolated components appeared to function together through substantially more of the session.

11. Current interpretation

Evidence of increasing **integration and persistence**, but insufficient evidence to declare complete restoration or automaticity.

12. Remaining uncertainties

- Repeatability next session.
- Whether ankle strategy remains consciously necessary.
- Whether transient adductor pinch continues.
- Whether proximal restrictions remain limiting at deeper hinge ROM.
- Durability under fatigue and progressive loading.

13. Related events

CF-15 through CF-19.

14. Confidence

High for reported observations; **Medium** for integration interpretation.

Part II — Exercise Evidence Records

Editorial note: Exercise Evidence Records are documented at the level of investigative classification and justification for each exercise, rather than the full chronological field structure used in Case Files and Failed Experiments. This reflects the form in which this evidence was originally captured.

ER-01 — Neutral Leg Press

Evidence Strength

Primary Discovery Exercise

Integration Exercise

Assessment Exercise

Development Exercise

Justification: Neutral Leg Press directly produced major evidence distinguishing **load acceptance from merely moving the weight**, demonstrated that subjective load distribution could deteriorate before absolute muscular failure, exposed the importance of

pelvic position, and eventually showed full-ROM acceptance extending through the pelvis into the oblique. It also repeatedly measured left-right distribution and persistence while simultaneously serving strengthening/hypertrophy and whole-chain integration.

ER-02 — High-Foot Leg Press

Evidence Strength

Major Supporting Exercise

Integration Exercise

Assessment Exercise

Development Exercise

Justification: High-Foot Leg Press repeatedly strengthened the developing load-acceptance model by showing that left-trunk participation could be deliberately maintained through the eccentric while remaining weak, and that **movement access and weakness were separable limitations**. Its evidence reinforced rather than originated the larger load-acceptance framework.

ER-03 — Horizontal Leg Press

Evidence Strength

Primary Discovery Exercise

Priming Exercise

Integration Exercise

Justification: Horizontal Leg Press directly generated the **proximal-versus-distal priming distinction** and, more importantly, initiated the late-August observation sequence that led to the multidirectional ankle-control discovery. Its very low-load environment allowed mechanics to become perceptible that conventional working loads had obscured. Its original and confirmed role remains proximal priming/integration.

ER-03A — Linear Leg Press

Evidence Strength

Major Supporting Exercise

Priming Exercise

Justification: Linear Leg Press helped establish that superficially similar presses could produce distinctly different effects. Its preferential knee/lower-leg response supplied the comparison necessary to strengthen the **selective priming model** developed alongside Horizontal Leg Press. It did not independently generate the subsequent ankle-control model.

ER-04 — Bulgarian Split Squat

Evidence Strength

Primary Discovery Exercise

Major Supporting Exercise

Integration Exercise

Assessment Exercise

Development Exercise

Justification: Bulgarian Split Squat directly produced major evidence that adductor symptoms could vary according to surrounding-chain organization and provided one of the clearest perceptions of **ground-to-hip force transmission**. It repeatedly exposed wobble, adductor behavior, glute/outer-ham participation and response to improved base mechanics while also serving unilateral hypertrophy/strength work.

ER-05 — Hamstring Curl

Evidence Strength

Major Supporting Exercise

Development Exercise

Assessment Exercise

Justification: Hamstring Curl became an important controlled environment for distinguishing **maximum movable load from optimal mechanical targeting**. It repeatedly strengthened evidence that preparatory contractions and load selection could change access to the proximal hamstring region. Its discoveries were significant, but primarily expanded the broader reintegration model rather than redirecting the overall investigation.

ER-06 — Back Extension: Hamstring Bias

Evidence Strength

Major Supporting Exercise

Integration Exercise

Development Exercise

Assessment Exercise

Justification: Hamstring-biased Back Extension repeatedly strengthened evidence that small positional changes materially alter posterior-chain recruitment and that increasing recruitment of previously deficient tissue can temporarily lower performance. It became an important second environment for accessing and assessing proximal hamstring participation.

ER-07 — Back Extension: Glute Bias

Evidence Strength

Confirmatory Exercise
Integration Exercise
Development Exercise
Assessment Exercise

Justification: Its strongest evidentiary contribution was independent confirmation that the proximal “strings” identified through Hamstring Curl could also be strongly accessed through a different movement pattern—hip extension rather than knee flexion. That cross-exercise reproduction strengthened the finding without originating it.

ER-08 — DB RDL / RDL

Evidence Strength

Major Supporting Exercise
Integration Exercise
Assessment Exercise
Development Exercise

Justification: RDL work has repeatedly exposed asymmetrical load distribution, left-trunk acceptance, grip limitations, lower-back substitution and the effect of residual hinge restriction. It has now become a high-resolution testing environment for foot, ankle, arch, hip and glute mechanics. Because the newer RDL mechanical investigation remains unresolved, it is not yet classified as the origin of a settled major discovery.

ER-09 — Standing Calf Raise

Evidence Strength

Major Supporting Exercise
Confirmatory Exercise
Integration Exercise
Assessment Exercise
Development Exercise

Justification: Standing Calf Raise evolved from calf development into a revealing test of distal-base organization. Most importantly, the August 28 session supplied **loaded confirmation** immediately after the ankle-control discovery: the ankle/arch strategy was applied and whole-leg participation was perceived. It therefore strongly supports and independently tests a discovery generated elsewhere rather than originating the ankle model itself.

ER-09A — Seated Calf Raise Experiment

Evidence Strength

Assessment Exercise

Development Exercise

Justification: The seated experiment primarily evaluated whether bent-knee calf work could provide a better or complementary stimulus while reducing unwanted limitations. The compromised bench geometry prevented the experiment from yielding sufficiently clean evidence to give it a stronger investigative classification.

ER-10 — Swimmers / Leg-Only Supermans

Evidence Strength

Priming Exercise

Integration Exercise

Justification: These movements primarily prepared the pelvic/posterior system for subsequent hamstring curls. Their main evidentiary value was demonstrating that preparatory hip/pelvic extension work could alter later knee-flexion recruitment, supporting a more integrated rather than isolated view of hamstring function.

ER-11 — Frog Glute Bridge

Evidence Strength

Integration Exercise

Priming Exercise

Major Supporting Exercise

Justification: Frog Glute Bridge began as glute activation but produced the important observation that anterior pelvic/top hip-flexor and posterior glute regions could be perceived functioning “**as one unit.**” It strengthened the broader pelvic-integration model while primarily serving pre-training incorporation rather than assessment or hypertrophy.

ER-12 — Hip Airplanes

Evidence Strength

Integration Exercise

Priming Exercise

Assessment Exercise

Major Supporting Exercise

Justification: Hip Airplanes repeatedly demonstrated that pelvic rotational organization materially affects left-side integration, particularly after the “Roman twist” cue improved pelvic draw-in. They became a reliable way to prepare and assess hip/pelvic organization,

but the record does not establish them as the origin of a discrete major working-model transition.

ER-13 — Pallof Press

Evidence Strength

Primary Discovery Exercise

Integration Exercise

Priming Exercise

Assessment Exercise

Development Exercise

Justification: Pallof Press produced one of the strongest immediate changes in single-leg balance observed during the investigation and materially expanded the model beyond the hip by demonstrating that trunk anti-rotation organization could influence the lower chain. Its role changed from generic “core” work to a central trunk/pelvis integration tool.

ER-14 — Left-Leg Reinforcement

Evidence Strength

Major Supporting Exercise

Priming Exercise

Integration Exercise

Assessment Exercise

Justification: Left-Leg Reinforcement repeatedly showed that low-load unilateral exposure could increase trunk, glute and outer-quad participation before bilateral pressing and appeared associated with reduced unwanted adductor loading. It strengthened multiple existing discoveries and evolved specifically into a pre-leg-press primer.

ER-15 — Single-Leg Weight-Acceptance Drill

Evidence Strength

Primary Discovery Exercise

Integration Exercise

Priming Exercise

Assessment Exercise

Justification: This stripped-down drill directly demonstrated that a movement could be organized specifically to teach **load acceptance itself**, independently of conventional hypertrophy or strength goals. That distinction is central to the working model and makes the drill evidentially more important than its informal/nonstandardized status might suggest.

ER-16 — Walking Lunges

Evidence Strength

Primary Discovery Exercise

Assessment Exercise

Development Exercise

Justification: Walking Lunges supplied important early evidence that proximal glute/hip tightening could precede or appear to provoke adductor symptoms. This materially broadened the earlier adductor-centered interpretation and helped establish that the adductor might sometimes be **downstream of another mechanical problem**. Dynamic unilateral loading also exposed deficits less evident under controlled bilateral conditions.

ER-17 — Roman Chair Lift

Evidence Strength

Major Supporting Exercise

Integration Exercise

Assessment Exercise

Development Exercise

Justification: Roman Chair work repeatedly strengthened evidence that the anterior pelvic/hip-flexor region was relevant to the larger chain rather than merely to abdominal performance. Foot spacing altered left-core participation, while later increases in execution difficulty and pelvic lift provided a continuing assessment of reincorporation.

ER-18 — Porch Standing / Ankle-Flexion Experiments

Evidence Strength

Primary Discovery Exercise

Assessment Exercise

Priming Exercise

Justification: Although not standardized as one formal exercise, the porch experiments were historically central investigative work. They directly produced an early ankle release, contributed to subsequent foot/ankle releases, and—most importantly—kept the unexplained **downward-pressure deficit** under active investigation long before its mechanism was understood. Later ankle findings retrospectively clarified what much of this experimentation had been probing. The Priming classification applies to portions of the porch work that were repeatedly used to prepare movement; the Assessment classification reflects repeated testing of whether downward pressure and ankle behavior had changed.

Exercise Evidence Classification Summary

Exercise	Evidence Strength	Primary Contribution	Supporting Discoveries	Current Role
ER-01 Neutral Leg Press	Primary Discovery; Integration; Assessment; Development	Established load acceptance as distinct from moving weight and exposed fatigue-related distribution loss	Pelvic positioning; eccentric participation; distal-to-proximal engagement	Strength/hypertrophy + reintegration + progress assessment
ER-02 High-Foot Leg Press	Major Supporting; Integration; Assessment; Development	Strengthened eccentric load-acceptance and weakness-vs-access distinctions	Trunk participation; trust/load acceptance	Posterior-chain development + reintegration/assessment
ER-03 Horizontal Leg Press	Primary Discovery; Priming; Integration	Generated proximal/distal priming distinction and initiated ankle-discovery sequence	Low-load mechanics; improved Bulgarian preparation	Active proximal primer; hypertrophy role still under study
ER-03A Linear Leg Press	Major Supporting; Priming	Established distinctly distal priming response	Selective priming according to chain need	Situational knee/lower-leg primer
ER-04 Bulgarian Split Squat	Primary Discovery; Major Supporting; Integration; Assessment	Exposed adductor-chain relationship and ground-to-hip force transmission	Outer-chain engagement; ankle/base effects	Unilateral hypertrophy + integration/assessment

Exercise	Evidence Strength	Primary Contribution	Supporting Discoveries	Current Role
	ent; Development			
ER-05 Hamstring Curl	Major Supporting; Development; Assessment	Distinguished target engagement from maximum load	Proximal-hamstring access; preparatory contraction effects	Hamstring development + targeting/assessment
ER-06 Ham-Bias Back Extension	Major Supporting; Integration; Development; Assessment	Strengthened proximal posterior-chain localization	Position-dependent recruitment; newly recruited tissue affecting reps	Posterior development + reintegration
ER-07 Glute-Bias Back Extension	Confirmatory; Integration; Development; Assessment	Independently confirmed proximal-region access through hip extension	Cross-exercise confirmation of “strings”	Glute/posterior development + confirmation
ER-08 DB RDL / RDL	Major Supporting; Integration; Assessment; Development	High-resolution test of hinge distribution and whole-chain mechanics	Foot/ankle organization; effective ROM; trunk acceptance	Active development and mechanical investigation
ER-09 Standing Calf Raise	Major Supporting; Confirm	Loaded confirmation of ankle/arch incorporation	Near-even distribution ; whole-chain	Calf development + distal-chain

Exercise	Evidence Strength atory; Integration; Assessment; Development	Primary Contribution	Supporting Discoveries n standing bracing	Current Role integration/assessment
ER-09A Seated Calf Raise	Assessment; Development	Tested alternate calf mechanics/setup	Bent- vs straight-knee stimulus distinction	Unresolved experimental alternative
ER-10 Swimmers / Leg-Only Supermans	Priming; Integration	Improved posterior/pelvic preparation before curls	Pelvic preparation affects hamstring access	Active warm-up/primer
ER-11 Frog Glute Bridge	Integration; Priming; Major Supporting	Linked anterior and posterior pelvic engagement	“One unit” pelvic/hip incorporation	Active pelvic/glute primer
ER-12 Hip Airplanes	Integration; Priming; Assessment; Major Supporting	Demonstrated importance of pelvic rotational organization	Pelvic draw-in; hip preparation	Active primer/reintegration assessment
ER-13 Pallof Press	Primary Discovery; Integration; Priming; Assessment; Development	Demonstrated trunk organization can materially alter lower-chain function	Single-leg balance; pelvis/trunk role in load acceptance	Active trunk/pelvis development and integration

Exercise	Evidence Strength	Primary Contribution	Supporting Discoveries	Current Role
ER-14 Left-Leg Reinforcement	Major Support ing; Priming; Integrat ion; Assessm ent	Demonstrated low-load pre-exposure can improve later bilateral participation	Outer quad/glute engagemen t; reduced adductor loading	Active pre-leg-press primer
ER-15 Single-Leg Weight-Acceptance Drill	Primary Discove ry; Integrat ion; Priming; Assessm ent	Isolated and taught load acceptance as its own capability	Distal bodyweight acceptance; repeated low-level practice	Active corrective/inte gration drill
ER-16 Walking Lunges	Primary Discove ry; Assessm ent; Develop ment	Helped overturn purely adductor-centere d interpretation	Proximal restriction preceding adductor symptoms	Historically diagnostic; development role status requires verification
ER-17 Roman Chair Lift	Major Support ing; Integrat ion; Assessm ent; Develop ment	Expanded abdominal work into anterior-pelvic/h ip-flexor evidence	Foot-spacin g effect; increased pelvic lift	Active trunk/anterior-pelvic development
ER-18 Porch Standing / Ankle-Flexion Experiments	Primary Discove ry; Assessm ent; Priming	Preserved and explored downward-press ure problem; produced early distal releases	Foot/ankle release; later ankle-contr ol reinterpret ation	Historical exploratory evidence; superseded by more deliberate testing

Editorial audit finding

The classification produces a useful distinction between **exercise importance to training** and **exercise importance to the investigation**.

The strongest **Primary Discovery Exercises** are:

Neutral Leg Press, Horizontal Leg Press, Bulgarian Split Squat, Pallof Press, Single-Leg Weight-Acceptance Drill, Walking Lunges, and Porch Standing/Ankle-Flexion Experiments.

They earned that status for different reasons. Some generated conceptual breakthroughs—such as **load acceptance** or the recognition that the problem extended into the trunk. Others exposed relationships that materially changed prior models—such as Walking Lunges challenging the adductor-centered interpretation or Horizontal Leg Press initiating the ankle-control investigation.

Several exercises that are extremely important to the current program are deliberately **not** classified as Primary Discovery Exercises. Standing Calf Raise, for example, became highly informative, but its strongest late-August contribution was **confirmatory**: it tested the ankle-control discovery under loaded standing conditions after that discovery had already emerged through Horizontal Leg Press, walking, and home mapping. Likewise, Glute-Bias Back Extension independently confirmed proximal-hamstring access that had first become clearly localized elsewhere.

This classification therefore preserves the historical distinction the case file is intended to capture:

where a discovery originated, where it was strengthened, where it was independently confirmed, and where it was eventually trained into the system.

Figure 7. Exercise Evidence Relationship Map

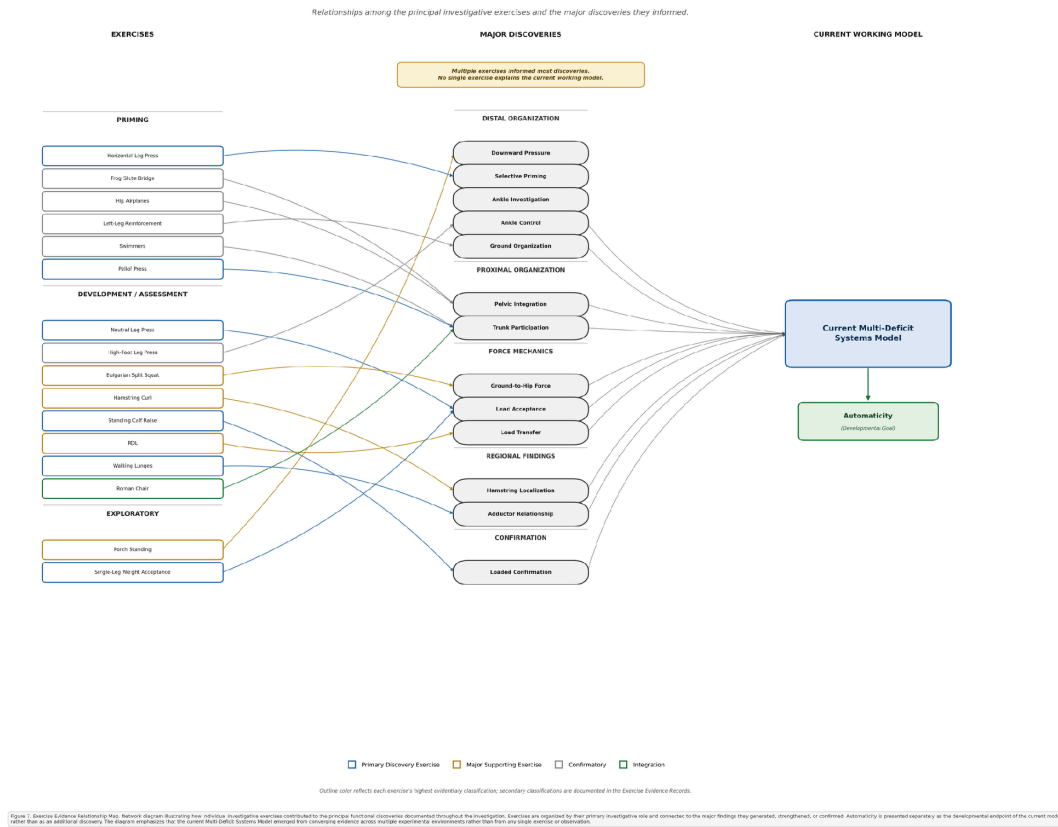


Figure 7. Exercise Evidence Relationship Map

5.7 Editorial Note

The records presented in this section constitute the evidentiary foundation upon which the remainder of this report is built.

The analytical sections that follow—including **Failed Experiments & Abandoned Hypotheses**, **Evolution of the Working Model**, and the **Current Multi-Deficit Systems Model**—should be understood as interpretations derived from these records rather than as independent sources of evidence.

Accordingly, readers seeking to evaluate the investigation are encouraged to consult the primary evidence before considering the analytical conclusions presented later in this report.

SECTION 6 — INVESTIGATIVE ANALYSIS

6.1 Purpose

The preceding sections presented the historical context, investigative methodology, chronology, and primary evidence accumulated throughout the investigation.

This section examines how that evidence altered the course of the investigation.

Unlike the Case Files and Exercise Evidence Records, which preserve observations in their investigative context, the present section evaluates the evolution of ideas. It documents how initial explanations emerged, how they were challenged by subsequent observations, and how the working model became progressively more comprehensive over time.

The objective is not to demonstrate that earlier interpretations were incorrect. Rather, it is to show how each successive stage of the investigation contributed to a more complete understanding of the functional problem.

Accordingly, this section distinguishes carefully between observations that remained stable throughout the investigation and the explanatory models that evolved in response to those observations.

6.2 Organization of the Analysis

The analytical record is presented in three complementary components.

Part I — Failed Experiments & Abandoned Hypotheses

This subsection documents experimental approaches, assumptions, and explanatory models that were ultimately revised, expanded, or abandoned.

Importantly, most entries do **not** represent failed observations.

Instead, they document situations in which observations remained valid while the explanations assigned to those observations became insufficient as additional evidence accumulated.

The progression documented here demonstrates the iterative nature of the investigative process.

Part II — Evolution of the Working Model

This subsection reconstructs the chronological development of the investigation's explanatory framework.

Each Working Model represents the best explanation supported by the available evidence at a particular stage of the investigation.

As new evidence accumulated, Working Models were expanded, refined, merged, or superseded according to the explanatory power of the developing evidence base.

The historical preservation of these models allows readers to understand not only the current explanation but also the investigative reasoning that produced it.

Part III — Current Working Model

The final analytical subsection presents the current multi-deficit systems model supported by the evidence assembled through Version 1.0 of this case file.

This model should be understood as the investigation's present explanatory framework rather than its final conclusion.

Because the investigation remains active, future evidence may refine, expand, or modify the current model.

6.3 Analytical Principles

Throughout this section, several editorial principles have been maintained.

Preservation of Historical Reasoning

Earlier interpretations are preserved as part of the historical investigation rather than rewritten according to later discoveries.

This allows readers to evaluate how understanding developed through time.

Separation of Observation and Explanation

Observations remain distinct from the explanations proposed to account for them.

Accordingly, an explanatory model may change while the underlying observations remain unchanged.

This distinction is particularly important throughout the evolution of the Working Models.

Progressive Refinement

The investigation demonstrated that understanding rarely advanced through complete replacement of previous ideas.

More commonly, earlier models proved to be:

- correct but incomplete;
- retained as components of larger explanations;
- refined through additional observations;
- or integrated into increasingly comprehensive frameworks.

Consequently, the analytical record emphasizes continuity of understanding rather than abrupt conceptual replacement.

Preservation of Uncertainty

Questions that remain unresolved are retained explicitly throughout the analytical record.

Uncertainty is treated as an essential characteristic of an active investigation rather than a deficiency to be eliminated through unsupported conclusions.

6.4 Relationship to the Evidence

The analytical conclusions presented throughout this section are derived directly from the primary evidence documented in **Section 5 — Investigative Evidence Records**.

Whenever possible, explanatory developments are supported by cross-references to:

- Case Files;
- Exercise Evidence Records;
- Chronological events;
- and previously established observations.

The analytical record therefore serves as an interpretation of the evidence rather than an independent source of evidence itself.

6.5 Analytical Sequence

The investigative analysis proceeds in the following order:

Chapter 6A

Failed Experiments & Abandoned Hypotheses



Chapter 6B

Evolution of the Working Model



Chapter 6C

Current Multi-Deficit Systems Model



Chapter 6D

Open Case Items & Future Investigation

This progression mirrors the development of the investigation itself:

failed explanations gave way to improved models, which ultimately converged upon the current working framework while leaving clearly defined questions for future investigation.

6.6 Editorial Note

Readers should resist interpreting the analytical sections as evidence that the investigation progressed through a series of incorrect ideas.

The historical record demonstrates a different pattern.

Most early observations proved to be valid.

What changed over time was not the evidence itself, but the explanatory framework required to account for that evidence.

Accordingly, the analytical development documented in this section should be understood as a process of increasing explanatory completeness rather than repeated conceptual replacement.

This distinction represents one of the defining methodological characteristics of the investigation and underlies the development of the current multi-deficit systems model.

Chapter 6A — Failed Experiments & Abandoned Hypotheses

Record status: Reconstructed from the available investigative history through August 28, 2026.

Classification rule: An intervention that worked but failed to explain the entire dysfunction is **not** classified as failed. The disposition reflects what happened to the *idea itself*, not merely whether the underlying left-chain problem persisted.

FE-01 — Original Adductor-Centered Explanation

1. Original hypothesis / assumption

The persistent left-side restriction was initially understood substantially through the prior **left adductor injury** and continuing adductor tightness/weakness.

2. Approximate chronology

- Origin: left adductor injury approximately 2025.
- Persisted into early 2026 investigation.
- Progressively revised during spring/summer 2026 as glute/hip, foot/ankle, knee, load-acceptance and trunk findings accumulated.

3. Initial supporting evidence

- Known acute injury occurred during adductor-machine use.
- Months of pain followed.

- Persistent tightness/weakness remained.
- Left knee-to-chest/hip flexion was restricted.
- Adductor pinching appeared during certain loaded movements.
- Walking lunges could produce increasing glute/hip tightness followed by adductor symptoms.
- Bulgarian split squats also intermittently produced an adductor pinch.

4. Evidence challenging/replacing it

- Improvement in the adductor did not restore the complete chain.
- Glute/hip restriction could apparently **trigger** the adductor rather than the adductor always being the initiating structure.
- Foot/ankle, push-off, knee stabilization, downward-pressure and trunk-acceptance deficits existed that the adductor injury alone could not adequately explain.
- By August 28, an adductor pinch could briefly appear and then recede when surrounding musculature re-engaged.

5. Final disposition

Partially retained / incorporated into broader working model.

The original injury remains relevant. The adductor is no longer treated as a sufficient explanation for the entire left-chain dysfunction.

6. Confidence

High.

7. Supporting record

- Original adductor-machine injury.
- Persistent knee-to-chest restriction.
- Walking-lunge/adductor observations.
- Bulgarian split-squat pinch observations.
- August 28 transient pinch followed by surrounding muscular re-engagement.

FE-02 — Treating the Left-Side Problem Primarily as a Proximal Glute/Hip Restriction

1. Original hypothesis / experiment

As the investigation moved beyond the adductor, substantial attention shifted toward the **glute/hip restriction**, eventually localized into two centers.

The implicit working possibility was that restoring those proximal areas might restore the chain.

2. Chronology

Spring 2026 → progressively revised through summer → materially expanded in late August.

3. Initial supporting evidence

- Hip/glute popping/release produced immediate functional changes.
- Left glute became fully squeezable after a major release.
- Hip hingement improved as proximal restriction decreased.
- Leg-press descent improved substantially.
- Massage, heat, stretching and proximal priming repeatedly changed distal movement quality.
- Training logs continued identifying glute/hip restriction even while left-trunk load acceptance improved.

4. Challenging/replacing evidence

Despite proximal improvements:

- push-off remained abnormal;
- downward pressure remained deficient;
- the foot couldn't adequately "grab" the ground;
- distal bracing remained incomplete;
- ankle/foot releases continued;
- knee/lower-leg behavior required separate work;
- later ankle-control experiments produced effects that could not be adequately attributed to proximal glute restriction.

The August ankle discovery therefore established that proximal restriction and distal-control failure were **different sub-issues**.

5. Final disposition

Incorporated into a broader working model.

Not abandoned. The two proximal centers remain active components, particularly regarding restricted hip hingement.

6. Confidence

High.

7. Supporting record

- Hip/glute release event.
- Two-center identification.
- Persistent hinge restriction.
- Improved glute squeeze following release.
- Continued distal deficits despite proximal improvement.
- August multidirectional ankle discovery.

FE-03 — “Release” as Sufficient Restoration

1. Original assumption

Major pops/releases initially carried the possibility that a mechanical restriction had finally been removed and normal function might follow.

2. Chronology

Spring 2026, especially during the hip/glute and subsequent ankle/foot release sequence.

3. Initial supporting evidence

- Individual pops sometimes produced immediate, substantial relief.
- Glute contraction changed dramatically after proximal release.
- Subsequent ankle/foot pops produced compounding relief.
- Arch acceptance improved.

4. Challenging evidence

Release repeatedly produced **new capability without complete automatic function**.

After tissue/joint-feeling restrictions diminished, the left side still required:

- deliberate load acceptance,
- strengthening,
- foot/ankle work,
- push-off retraining,
- knee/thigh stabilization,
- pelvic/trunk integration,
- and eventually deliberate multidirectional ankle control.

Thus:

mobility/access ≠ incorporation ≠ automaticity.

5. Final disposition

Superseded as a sufficient explanation; partially retained as an intervention.

Release remains useful. The idea that release itself completes restoration does not.

6. Confidence

High.

7. Supporting record

- Hip/glute pop followed by restored squeeze but persistent outer-hip resistance.
- Sequential ankle/foot pops followed by improvement but continuing deficits.
- Subsequent requirement for extensive load-acceptance and integration work.

FE-04 — Individual Muscle Engagement as Equivalent to Chain Restoration

1. Original working assumption

Early discoveries understandably focused on getting particular structures to engage: glute, outer thigh, hamstring, arch, etc.

2. Chronology

Spring through summer 2026; progressively superseded by whole-chain observations.

3. Initial supporting evidence

- Previously inaccessible muscles became voluntarily contractible.
- Individual engagement changes immediately improved exercises.
- Outer-leg engagement could reduce adductor pinching.
- Proximal hamstring targeting became increasingly precise.

4. Challenging evidence

A muscle could be accessible while the chain remained deficient.

Examples:

- Glute became fully squeezable while other left-chain problems persisted.
- Arch improved while downward-pressure capability remained deficient.
- Left trunk accepted more load while weakness/glute-hip restriction remained.
- Different components needed to cooperate simultaneously before normal force transfer appeared.

5. Final disposition

Incorporated into broader working model.

Muscle engagement remains necessary but is no longer considered equivalent to restoration.

This contributed to the emerging sequence:

engagement → neurological connection → trust → load acceptance → load transfer.

6. Confidence

High.

7. Supporting record

- Post-pop glute squeeze.
- Arch “owns the ground.”
- Progressive trunk-load acceptance.
- August 28 integrated-session observations.

FE-05 — Arch/Foot Contact as a Complete Explanation for the Base

1. Original hypothesis

As the arch began to “own the ground,” improved foot contact appeared to represent restoration of the distal foundation.

2. Chronology

Spring/summer 2026 → substantially revised in late August.

3. Initial supporting evidence

- Foot/ankle pops produced relief.
- Arch acceptance improved.
- Left foot began attempting more normal push-off.
- Ground contact became perceptibly stronger.

4. Challenging evidence

Even with improved arch contact:

- downward pressure remained problematic;
- the left side still couldn't brace at the base like the right;
- “grabbing” the ground during walking remained incomplete.

The late-August discovery distinguished **contact with the ground** from **muscular organization around the ankle against the ground**.

The right side demonstrated a much fuller multidirectional/“360°” ankle-bracing capability.

5. Final disposition

Partially retained / incorporated into broader model.

Arch contact is important but insufficient by itself.

6. Confidence

High.

7. Supporting record

- Foot/ankle release sequence.
- “Arch owns the ground” observation.
- Persistent downward-pressure deficit.
- August multidirectional ankle-control mapping.

FE-06 — Foot Structural Tightness and Ankle Muscular Dysfunction Treated as One Problem

1. Original working model

Earlier foot/ankle difficulties were not yet divided cleanly into separate structural/release and muscular-control components.

2. Chronology

Spring–summer 2026 → differentiated August 2026.

3. Initial supporting evidence

- Foot cramping.
- Tight-feeling bones/areas.
- Toe-crossing behavior.
- Ankle/foot popping.
- Relief following pops.
- Improved arch behavior afterward.

These observations naturally grouped “foot/ankle” together.

4. Evidence producing revision

The later experiments demonstrated another phenomenon:

muscular exertion at the ankle itself could alter engagement farther up the chain even without another structural release occurring.

User therefore explicitly distinguished:

- remaining foot areas that feel tight and need to loosen/crack;

- ankle muscular-control deficiency.

5. Final disposition

Original combined model superseded by a two-component model.

6. Confidence

High for the conceptual distinction; exact anatomy remains uncertain.

7. Supporting record

- Repeated foot/ankle releases.
- August 28 discussion distinguishing foot tightness from ankle muscular control.

FE-07 — Downward-Pressure Deficit Without an Identified Mechanism

1. Original hypothesis / unresolved observation

The left side could not properly apply **downward pressure**, but the reason was unknown.

This is important: this was **not a failed observation**. The observation remained stable while its explanation changed.

2. Chronology

Recognized well before August 2026 → remained an investigative driver → partially explained August 2026.

3. Initial supporting evidence

- Repeated inability to push downward normally.
- Poor ground-grabbing during walking.
- Poor push-off.
- Left side did not brace against the ground like right.
- Motivated porch standing, ankle flexing and numerous other experiments.

4. Replacement evidence

Horizontal leg-press priming produced a clue.

A later walk exposed another relationship.

Home testing then mapped ankle exertion in multiple directions.

Comparison with the right demonstrated much fuller 360° ankle control.

Applying the newly discovered strategy during August 28 training changed chain engagement.

5. Final disposition

Still partially unresolved, with a substantially improved working explanation.

The ankle muscular-control deficit now appears to explain an important portion of the inability to generate downward pressure, but the case does **not** establish that it explains every component.

6. Confidence

High for original deficit; **Medium** for current mechanistic interpretation.

7. Supporting record

- Longstanding downward-pressure observation.
- Porch experimentation.
- Horizontal press discovery.
- Walking observation.
- Home ankle mapping.
- August 28 loaded application.

FE-08 — “Drive Through the Heel” as a Sufficient Foot Cue

1. Original experiment/cue

Heel-driving was used in lower-body work, including leg-press experimentation.

2. Chronology

Earlier lower-body reintegration → progressively refined through July/August.

3. Initial supporting evidence

- Heel/back-of-knee cues improved some movements.
- Heel pressure could help access desired posterior/proximal structures.

4. Challenging evidence

Later discoveries showed that force organization was more complex than heel pressure alone:

- pressing into the pad while simultaneously driving/sliding the foot downward inside the shoe altered anterior/medial-chain engagement;
- arch participation mattered;
- ankle multidirectional control mattered;
- during the August 28 Bulgarian split squat, the relevant observation became **force traveling from/through the heel to/through the hip**, rather than simply “push through heel.”

5. Final disposition

Partially retained.

Heel involvement remains useful, but heel-only thinking is incomplete.

6. Confidence

High.

7. Supporting record

- Earlier heel/back-of-knee cues.
- July leg-press foot-down cue.
- August ankle discovery.
- August 28 Bulgarian observation.

FE-09 — Leg-Press Loading Without the “Foot Down Inside the Shoe” Component

1. Original experiment

Conventional pressing into the leg-press platform.

2. Chronology

Before July 2026 → revised during July experimentation.

3. Initial supporting evidence

Leg press was already producing increasing left-trunk load acceptance and bilateral distribution improvements.

4. Challenging evidence

User discovered that simultaneously:

- pressing **into** the pad, and
- pressing/sliding the foot **down inside the shoe** without actually moving the shoe

changed engagement through the knee/inner thigh/anterior chain and helped maintain it during the eccentric.

5. Final disposition

Superseded by a more complete cue.

The exercise itself was not failed; the original force application was incomplete.

6. Confidence

High.

7. Supporting record

- Leg-press log observations.
- July 7 foot-pressure discovery.

FE-10 — Increasing Load When Engagement Was Not Preserved

1. Original experiment

Testing heavier loads during reintegration exercises.

2. Chronology

Repeated throughout lower-body rehabilitation/reintegration.

3. Initial supporting evidence

Higher loads were physically movable.

4. Challenging evidence

On multiple exercises, increased load could degrade the exact muscular participation being investigated.

Most clearly:

- Bulgarian split squat weight was reduced when engagement could no longer be maintained.
- Hamstring curl could be performed at **40 lb**, yet **30 lb** was identified as the sweet spot for the desired proximal hamstring “strings.”

5. Final disposition

Abandoned as a progression criterion.

Important qualification: the user did **not** hold a general belief that “more weight = more progress.” Rather, heavier-load trials served as tests, and loads were rejected when mechanics deteriorated.

6. Confidence

High.

7. Supporting record

- Bulgarian load reduction.
- August 28 hamstring-curl 30-lb sweet spot versus 40-lb capability.
- Repeated technical-form termination standard.

FE-11 — Standing Calf Raise as a Straightforward Calf-Only Exercise

1. Original working use

Standing calf raise used conventionally as calf training.

2. Chronology

Earlier program → challenged during left-chain investigation → reconsidered summer 2026 → newly reinterpreted August 28.

3. Initial supporting evidence

Standing machine allowed progressive calf loading.

4. Challenging evidence

- Left Achilles/lower-leg region could fatigue before the calf.
- This prompted investigation of seated calf alternatives.
- Later ankle/foot discoveries revealed the standing calf raise could expose/require much more extensive chain organization.
- August 28: whole left leg trembled on each rep/set while the ankle/arch and larger chain were being deliberately incorporated.

5. Final disposition

Incorporated into broader model.

Standing calf raise remains a calf exercise but has become diagnostically/integratively informative for the distal chain.

6. Confidence

High.

7. Supporting record

- Achilles/lower-leg fatigue observations.
- seated-calf experimentation.
- August 28 standing-calf note.

FE-12 — Seated Dumbbell Calf Raise as the Obvious Replacement

1. Original experiment

Replace standing calf raises with seated dumbbell calf raises to reduce unwanted Achilles/lower-chain limitation and alter calf emphasis.

2. Chronology

Summer 2026.

3. Initial supporting evidence

- Bent-knee calf work was theoretically attractive.
- User perceived more outer-calf involvement.
- It potentially offered another ankle/calf stimulus.

4. Challenging evidence

Gym benches were too low.

This placed knees higher than hips and produced an angled rather than satisfactorily vertical loading path.

Initial straight- versus bent-knee testing also produced similar unwanted lower-chain sensation.

5. Final disposition

Still unresolved / not adopted as replacement.

Standing machine remains available while a mechanically satisfactory seated setup has not been established.

6. Confidence

High.

7. Supporting record

- Seated DB calf experimentation.
- bench-height observation.
- straight/bent-knee comparison.

FE-13 — Stacking Horizontal and Linear Leg Press Because Both Help

1. Original possibility

Because horizontal and linear presses each produced useful priming effects, using both could potentially produce greater benefit.

2. Chronology

August 2026.

3. Initial supporting evidence

- Horizontal press strongly affected proximal chain.
- Linear press strongly affected knee/lower-leg region.

4. Evidence refining the idea

Their benefits were discovered to be **different rather than simply additive**.

Horizontal:

proximal primer.

Linear:

distal primer.

No evidence established that routinely stacking both was superior to selecting according to current need.

5. Final disposition

Not adopted as a general rule. Partially retained as situational options.

6. Confidence

Medium-High.

7. Supporting record

- Comparative horizontal/linear leg-press testing.

FE-14 — Horizontal Leg Press as Merely a Primer

1. Original model

Horizontal leg press was identified as an unusually effective **proximal primer**.

2. Chronology

August 2026 → subsequently reopened.

3. Initial supporting evidence

Very low loads and partial/pulse repetitions produced strong access to:

- glute,
- pelvic/hip-flexor region,
- trunk of thigh,
- outer quad/knee,
- trunk engagement.

4. Evidence challenging the limited classification

The response was sufficiently useful that the user began considering whether heavier/high-foot horizontal pressing could preserve the desired pathway while also providing a legitimate hypertrophy stimulus.

5. Final disposition

Still unresolved.

The priming role is retained. A second role as a main hypertrophy movement remains under investigation.

6. Confidence

High for priming role; **Low-Medium** for hypertrophy role.

7. Supporting record

- Two successful horizontal-leg-press priming tests.
- Subsequent discussion of possible main-routine use.

FE-15 — Leg Extensions as Appropriate During Early Reintegration

1. Original programming element

Leg extensions were part of lower-body training.

2. Chronology

Earlier programming → removed during left-chain reCOORDinating period → rationale now being reconsidered as function improves.

3. Initial supporting evidence

Leg extensions provide direct quadriceps work.

4. Evidence challenging their continued use

During the reintegration period, isolated knee extension did not provide the integrated loading environment needed to coordinate:

- foot,
- knee,
- thigh,
- glute,
- pelvis,
- trunk.

Integrated leg-press loading was prioritized instead.

5. Final disposition

Temporarily abandoned; current exclusion may be approaching expiration.

This is not evidence that leg extensions are inherently inappropriate. They were removed because they did not serve the specific reintegration priority at that stage.

6. Confidence

High.

7. Supporting record

- Earlier routine.
- later removal.
- subsequent discussion of why they were removed and whether reintroduction is now appropriate.

FE-16 — Adductor Pinch as a Signal That the Movement Must Necessarily Stop

1. Original practical interpretation

Adductor pinching had historically represented a meaningful movement limitation.

2. Chronology

Earlier lower-body sessions → revised during successful Bulgarian work in August.

3. Initial supporting evidence

- Pinch accompanied restriction.
- Walking lunges had been terminated when hip/glute tightening triggered adductor symptoms.
- Bulgarian split squats had previously produced pinching.

4. Challenging evidence

During later Bulgarian work:

- pinch appeared;
- maintaining/re-establishing outer-leg engagement caused it to recede.

August 28 produced an even more notable event:

- brief pinch appeared;
- surrounding muscles re-engaged;
- pinch disappeared.

5. Final disposition

Partially revised.

Pinching remains relevant feedback and is not to be ignored, but its appearance no longer necessarily means the entire movement pattern has failed irreversibly. Under some circumstances the desired organization can be restored.

6. Confidence

Medium-High.

7. Supporting record

- Walking-lunge termination.
- Bulgarian pinch observations.
- August 28 spontaneous/rapid re-engagement.

FE-17 — Conscious Engagement as the End Goal

1. Original investigative necessity

Early restoration depended heavily on consciously finding and activating individual structures.

2. Chronology

Throughout 2026 → increasingly revised as access improves.

3. Initial supporting evidence

Deliberate cues repeatedly produced immediate improvements.

4. Challenging evidence

The user has progressively distinguished:

- being able to find something,
- being able to incorporate it,
- maintaining it under load,
- maintaining it under fatigue,
- and having the nervous system select it automatically.

The right ankle now provides a particularly clear control: its multidirectional bracing occurs automatically, whereas the left currently requires more conscious organization.

5. Final disposition

Superseded as the endpoint.

Conscious engagement remains a developmental tool. **Automaticity** is the intended endpoint.

6. Confidence

High.

7. Supporting record

- Repeated reintegration work.
- 95–98% session-long load-acceptance estimate.
- August ankle right/left comparison.
- explicit current goal of making ankle behavior automatic.

FE-18 — One Unified Cause for the “Offline Chain”

1. Original tendency of the investigation

At different stages, individual discoveries naturally became candidates for explaining a large portion of the dysfunction:

- adductor,
- proximal glute/hip restriction,
- foot/arch restriction,
- knee/thigh behavior,
- ankle.

2. Chronology

2025–August 2026.

3. Initial supporting evidence

Each discovery genuinely explained **some previously unexplained observations** and often produced immediate functional improvement.

4. Evidence replacing the single-cause model

No single intervention restored all functions.

By August 28, the user explicitly articulated the revised model:

There are **several sub-issues**, affecting different functions, that converge on the larger condition of the **left chain being offline**.

Examples:

- two centers → hip-hinge restriction;
- foot tightness → continuing distal restriction/release phenomena;
- ankle muscular deficit → base bracing/ground grabbing/downward pressure;
- knee/thigh deficits → step stabilization/load acceptance;
- glute/pelvis/trunk deficits → proximal acceptance/transfer.

5. Final disposition

Abandoned as a single-cause framework. Replaced by multi-deficit systems model.

6. Confidence

High.

7. Supporting record

The entire longitudinal sequence, culminating in the August 28 clarification that the ankle explains a subset of the dysfunction rather than the original deficit as a whole.

FE-19 — “The Foot Is Planted” as Sufficient Information About Foot Mechanics

1. Original assumption

A stable/planted foot could initially be treated as the relevant base condition during exercises such as RDLs.

2. Chronology

Implicit earlier assumption → explicitly questioned August 2026.

3. Initial supporting evidence

Standard exercise mechanics emphasize maintaining stable ground contact.

4. Challenging evidence

User's investigation progressively distinguished:

- heel pressure,
- arch participation,
- forefoot/pad participation,
- push-off,
- downward pressure,
- ankle directional exertion,
- and muscular bracing against the ground.

The recent RDL discussion specifically questioned whether:

- toes should grip or spread,
- short-foot should be used,
- heel/pad/arch should be loaded independently or together.

The subsequent ankle discovery demonstrated why “keep the foot planted” is too low-resolution for this investigation.

5. Final disposition

Superseded by a more detailed foot/ankle force-management model.

Exact optimal RDL implementation remains unresolved.

6. Confidence

High that the simpler model is incomplete; **Low-Medium** regarding final RDL mechanics.

7. Supporting record

- RDL foot-mechanics discussion.
- prior arch discoveries.
- August multidirectional ankle mapping.

FE-20 — Predetermined RDL ROM

1. Original issue under investigation

Whether RDL ROM should be determined externally by conventional depth landmarks versus internally by preservation of the desired hip/glute/chain mechanics.

2. Chronology

Explicitly investigated August 2026.

3. Initial supporting basis

Conventional RDL instruction commonly provides visual depth/position cues.

4. Evidence challenging a fixed-ROM model

User identified that whether the glute/flexor region remains actively organized or relaxes during the eccentric materially changes available ROM.

This made ROM itself part of the mechanical experiment rather than an independent target.

5. Final disposition

Still unresolved experimentally.

Current working direction: useful ROM should correspond to the range in which the desired hinge and force relationships remain intact, but the user's precise foot/ankle/hip strategy is still being mapped.

6. Confidence

Medium.

7. Supporting record

- August RDL mechanics discussion immediately preceding ankle-control discovery.

FE-21 — Left/Right Load Distribution as a Simple Strength Problem

1. Original possible interpretation

The left side's inability to take an equal share of bilateral load could be interpreted primarily as weakness.

2. Chronology

Early reintegration → progressively revised.

3. Initial supporting evidence

- Left side was objectively/functionally weaker.
- Load-distribution gap increased with fatigue.
- Training logs noted weakness limiting performance.

4. Challenging evidence

Distribution improved through **mechanical changes without simply becoming stronger first**:

- pelvic positioning,
- deliberate load acceptance,
- foot-pressure changes,
- outer-leg engagement,
- arch participation,
- ankle control,

- trunk incorporation.

The log progressed toward approximately **45:55** leg-press distribution and near-even calf distribution even while weakness/restriction remained.

5. Final disposition

Partially retained / incorporated into broader model.

Weakness is real, but weakness alone does not explain the asymmetry.

6. Confidence

High.

7. Supporting record

- workout-log load-distribution observations;
- fatigue widening the asymmetry;
- immediate improvements from mechanical cues;
- August 28 session-level load-acceptance observation.

Investigative Disposition

The strongest pattern in this section is **not a long trail of failed ideas**.

The record instead contains relatively few outright failures and a much larger number of **correct observations that initially had explanations that were too narrow**.

The major progression has been:

isolated injury → isolated restriction → individual muscle access → regional integration → load acceptance → force transfer → distal/proximal coordination → multi-deficit chain model → automaticity.

That distinction should be preserved when the eventual monograph is assembled. It would materially misrepresent the investigation to describe the earlier hypotheses as simply “wrong.”

Significant Evidentiary Gaps Requiring Possible Interview

I can identify several gaps that the available history does **not** resolve confidently enough to enter as findings.

1. **Earliest downward-pressure observation:** I can establish that it predates the recent ankle discovery and that it drove considerable experimentation, but I cannot establish exactly when you first recognized it or what activity first exposed it.
2. **Evolution of the “two centers” model:** The current record establishes the two centers and their characteristics, but not enough of the chronology showing whether

you discovered them simultaneously or sequentially, or what specifically caused you to distinguish them.

3. **Early competing anatomical hypotheses:** There are references to possible IT-band/vastus-lateralis involvement and possible quadratus-femoris/ischial-region involvement, but the available record isn't sufficient for me to determine which of those you genuinely adopted as working hypotheses versus anatomical possibilities we merely discussed. I therefore did **not** classify them as abandoned hypotheses.
4. **Foot-cracking hypotheses:** The chronology establishes repeated popping/release and relief, but not whether you held specific hypotheses about *which bones/joints* were bound before later discoveries revised those ideas.
5. **Pelvic-tilt relationship:** You've been working on a relatively minor pelvic tilt and believe the integrated chain work may help it, but the record doesn't establish whether pelvic tilt was ever considered a **cause** of the left-chain dysfunction. I therefore did not classify it as a failed or superseded hypothesis.
6. **Pallof press mechanism:** Pallof clearly worked and remains retained. What is missing is whether your explanation for *why* it worked changed materially enough to qualify for this section.
7. **Hip-airplane/Roman-twist mechanism:** Same issue: successful intervention, but insufficient evidence of an abandoned explanatory model.
8. **Horizontal-leg-press → ankle discovery:** We know the chronological sequence—horizontal press, later walking observation, home mapping, August 28 loaded application—but not yet the exact sensation during that first horizontal-press session that planted the ankle clue.
9. **Ankle directional map:** You have established that up/down/left/right exertion corresponds to different areas of glute/trunk engagement, but the current record doesn't contain the actual directional map. Therefore it cannot yet be audited for repeatability or later revision.
10. **Other genuinely failed porch experiments:** We have extensive evidence that you tried many movements while searching for the downward-pressure mechanism. The record disproportionately preserves the things that **worked**. It almost certainly undercaptures experiments that produced nothing and were discarded.

Those are the places where an interview could add genuinely new evidence rather than making you repeat information you've already supplied. I would **not interview you about the other entries above unless you identify an error or want greater chronological precision**.

Chapter 6B — Evolution of the Working Model

Editorial note: as with the Exercise Evidence Records, Working Model entries are documented here as title, Model Lineage, and explanatory reasoning, rather than the full chronological

field structure used in Case Files and Failed Experiments. This reflects the form in which this evidence was originally captured.

Figure 2. Evolution of the Working Model

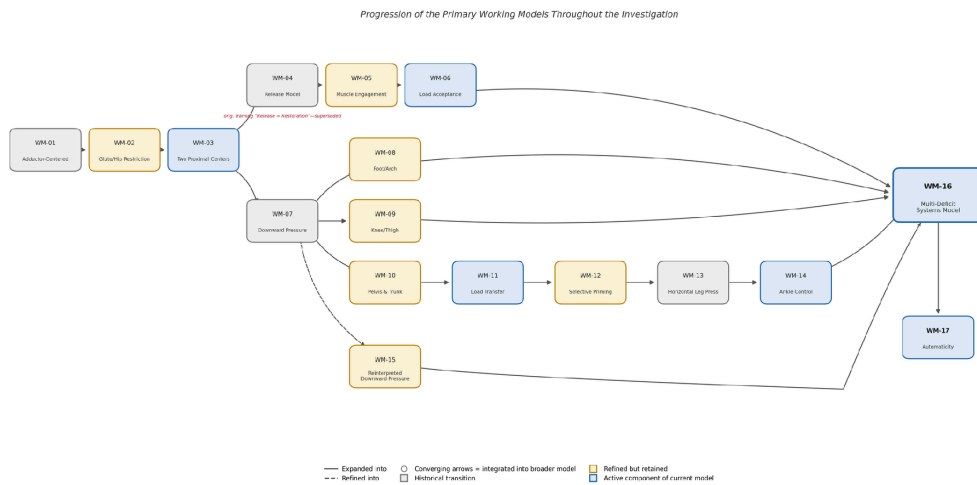


Figure 2. Evolution of the Working Model. Sequential development of the primary working models (WM-01 to WM-17) documented during the investigation. The figure illustrates how individual explanatory models were progressively expanded, refined, merged, or incorporated into later models rather than being simply discarded. Converging pathways culminate in the current Multi-Direct Systems Model (WM-16), while the developmental objective of the investigation is represented separately by the Automaticity model (WM-17).

Figure 2. Evolution of the Working Model

WM-01 — Local Injury / Adductor-Centered Model

Model Lineage

Introduced in: WM-01

Expanded into: WM-02

Partially retained in: WM-16

Current descendant: WM-16

WM-02 — Glute/Hip Restriction Becomes the Dominant Proximal Model

Model Lineage

Expanded from: WM-01

Refined into: WM-03

Continues as part of: WM-16

Current descendant: WM-16

WM-03 — Two-Center Proximal Restriction Model

Model Lineage

Refined from: WM-02

Continues as part of: WM-16

Current descendant: WM-16

The important lineage point here is that WM-03 was **not superseded** by the later distal discoveries. The completed section explicitly preserves it as the proximal hinge-restriction branch of the larger model.

WM-04 — Release Appears to Be the Key to Restoration

Model Lineage

Introduced in: WM-04

Refined into: WM-05

Partially retained in: WM-16

Superseded by: WM-05, specifically as a sufficient restoration model

Current descendant: WM-16

The **release intervention** survived; **release = restoration** did not. That distinction is necessary to keep the lineage accurate.

WM-05 — Individual Muscle Re-Engagement Model

Model Lineage

Refined from: WM-04

Expanded into: WM-06

Continues as part of: WM-06

Continues as part of: WM-16

Current descendant: WM-16 and WM-17

WM-05 survives particularly clearly inside the functional hierarchy preserved later: **muscle engagement** → **neurological connection** → **trust** → **load acceptance** → **load transfer**.

WM-06 — Load Acceptance Emerges as a Central Concept

Model Lineage

Expanded from: WM-05

Expanded into: WM-11

Continues as part of: WM-16

Continues as part of: WM-17

Current descendant: WM-16 and WM-17

WM-06 is one of the principal models that **survives rather than gets replaced**. WM-11 adds load transfer above it; WM-17 adds the developmental question of automaticity.

WM-07 — Downward Pressure and Push-Off Become Distinct Unresolved Functions

Model Lineage

Introduced in: WM-07

Expanded into: WM-08 and WM-09

Refined into: WM-13

Refined into: WM-15

Continues as part of: WM-16

Current descendant: WM-15 and WM-16

This is one of the clearest investigative lineages: the **observation survived unchanged**, while its explanation became progressively better. The completed record explicitly describes downward pressure as an investigative anchor whose mechanism remained unresolved until the ankle investigation.

WM-08 — Foot/Arch Becomes Part of the Chain

Model Lineage

Expanded from: WM-07

Refined into: WM-13

Expanded into: WM-14

Continues as part of: WM-16

Current descendant: WM-16

WM-14 does not invalidate WM-08. Instead, it separates the earlier broad foot/ankle conception into **structural/contact behavior** and **active ankle muscular control**.

WM-09 — Knee/Thigh Stabilization and the Missing “Catch”

Model Lineage

Expanded from: WM-07

Merged into: WM-16

Continues as part of: WM-16

Current descendant: WM-16

No evidence in the completed section supports calling WM-09 superseded. The knee/thigh stabilization branch remains explicitly present in WM-16.

WM-10 — Pelvis and Trunk Enter the Model as Active Load-Bearing Components

Model Lineage

Introduced in: WM-10

Expanded into: WM-11

Continues as part of: WM-16

Continues as part of: WM-17

Current descendant: WM-16 and WM-17

WM-10 changes the geographical boundary of the chain itself: the model stops at neither the foot nor the hip; the ipsilateral pelvis/trunk becomes part of the load pathway.

WM-11 — Load Transfer Becomes Distinct from Load Acceptance

Model Lineage

Expanded from: WM-06 and WM-10

Continues as part of: WM-16

Continues as part of: WM-17

Current descendant: WM-16 and WM-17

WM-11 does not supersede WM-06. It establishes the next higher-order functional question: **after accepting load, can the reconstructed system transmit force?**

WM-12 — Priming Becomes Selective Rather Than Generic

Model Lineage

Introduced in: WM-12

Expanded into: WM-13

Continues as part of: WM-16

WM-12 remains an intervention model rather than becoming the principal explanatory model. Its selective-priming logic directly provides the experimental environment from which WM-13 emerges.

WM-13 — Horizontal Leg Press Opens the Ankle Investigation

Model Lineage

Expanded from: WM-08 and WM-12

Refined from: WM-07

Expanded into: WM-14

Current descendant: WM-14, WM-15 and WM-16

This is principally a **transition model**: the unresolved base problem meets a selective experimental environment and redirects investigation from foot contact/pressure toward active ankle control.

WM-14 — Multidirectional Ankle-Control Model

Model Lineage

Expanded from: WM-13

Refined into: WM-15

Continues as part of: WM-16

Continues as part of: WM-17

Current descendant: WM-16 and WM-17

WM-14 also preserves rather than replaces the structural-foot component of WM-08 and the proximal two-center component of WM-03. That preservation is what eventually necessitates WM-16.

WM-15 — Retrospective Reinterpretation of the Downward-Pressure Problem

Model Lineage

Refined from: WM-07 and WM-14

Merged into: WM-16

Continues as part of: WM-16

Current descendant: WM-16

This lineage is unusual because WM-15 revises the **explanation of an old observation**, not the observation itself: downward-pressure failure goes from unexplained function to a function substantially constrained by deficient ankle/base organization, while other deficits remain operative.

WM-16 — Current Multi-Deficit Systems Model Emerges

Model Lineage

Merged from: WM-01, WM-02, WM-03, WM-05, WM-06, WM-07, WM-08, WM-09, WM-10, WM-11, WM-14 and WM-15

Partially retains: WM-04

Continues alongside: WM-12

Expanded into: WM-17

Current descendant: WM-16 and WM-17

This is the principal **convergence point**. Earlier valid models become branches or functional principles within a multi-deficit system rather than competing explanations. WM-04 is only partially retained because release remains useful while release-as-restoration was rejected.

WM-17 — Automaticity Becomes the Endpoint

Model Lineage

Expanded from: WM-05, WM-06, WM-10, WM-11, WM-14 and WM-16

Continues as part of: Current working model

Current descendant: WM-17

WM-17 does **not** replace WM-16. The completed section explicitly says the two frameworks describe different dimensions of the same investigation. WM-16 explains **what components constitute the present system**; WM-17 explains **the developmental state toward which those components must progress**.

Working Model Evolution Map

Working Model	Direct Parent	Direct Successor(s)	Final Disposition	Present Status
WM-01	—	WM-02	Partially retained / incorporated	Historical origin; survives

Working Model	Direct Parent	Direct Successor(s)	Final Disposition	Present Status
				within multi-deficit model
WM-02	WM-01	WM-03	Correct but incomplete / retained	Active proximal principle within WM-16
WM-03	WM-02	WM-16	Retained / incorporated	Active proximal hinge-restriction branch
WM-04	—	WM-05	Release-as-restoration superseded; release itself retained	Intervention principle only
WM-05	WM-04	WM-06	Incorporated	Engagement remains necessary precursor
WM-06	WM-05	WM-11	Major retained framework	Active functional hierarchy
WM-07	—	WM-08, WM-09, WM-13, WM-15	Observation retained; explanation progressively refined	Downward pressure/push-off remain tracked functions
WM-08	WM-07	WM-13, WM-14	Correct but incomplete / incorporated	Foot structural/contact branch remains active
WM-09	WM-07	WM-16	Retained as component	Active knee/thigh stabilization branch
WM-10	—	WM-11	Major retained expansion	Active pelvis/trunk branch
WM-11	WM-06, WM-10	WM-16, WM-17	Retained	Load transfer remains higher-order

Working Model	Direct Parent	Direct Successor(s)	Final Disposition	Present Status
WM-12	—	WM-13	Retained/refined	functional outcome Active selective-priming framework
WM-13	WM-07, WM-08, WM-12	WM-14	Major transition	Historical bridge into ankle-control model
WM-14	WM-13	WM-15, WM-16, WM-17	Active major component	Active ankle-control branch
WM-15	WM-07, WM-14	WM-16	Incorporated	Current explanation of downward-pressure deficit
WM-16	Multiple converging prior models	WM-17	Current primary working model	Active
WM-17	WM-16 plus retained functional frameworks	—	Current endpoint model	Active

Figure 3. Working Model Lineage Diagram

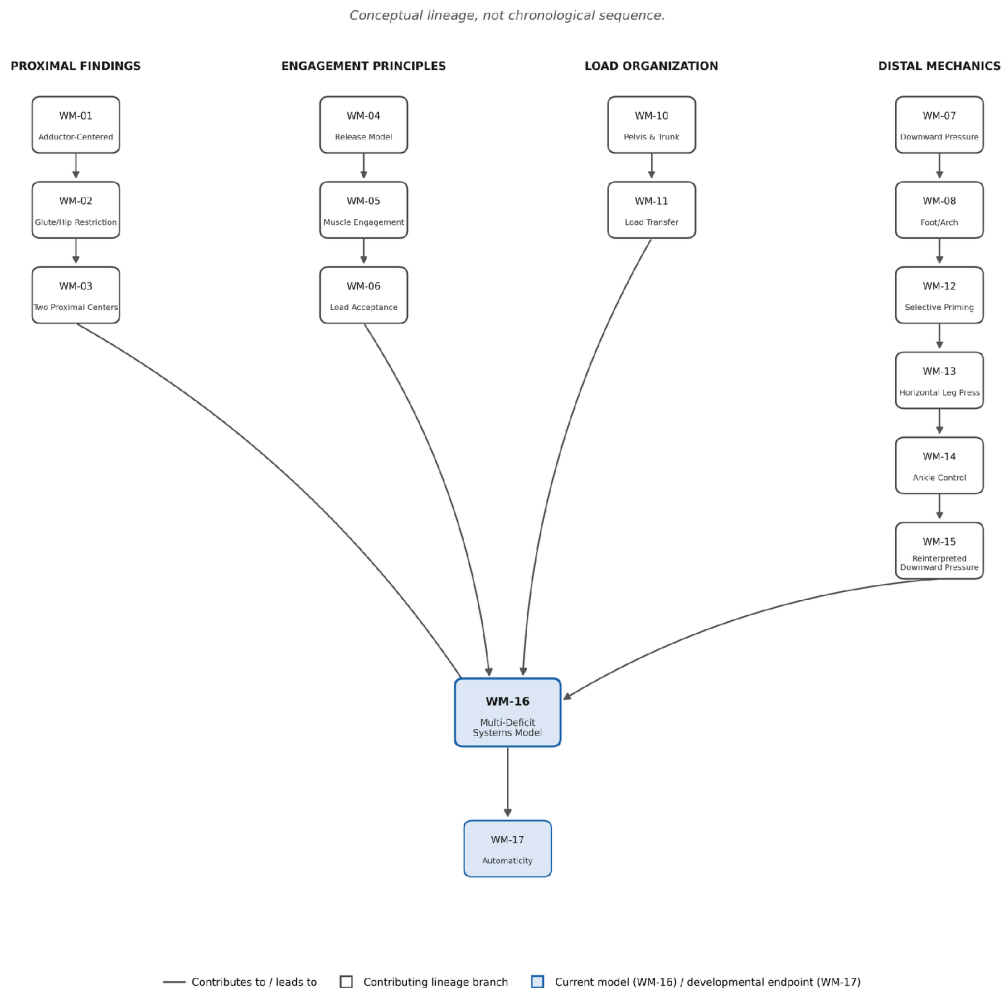


Figure 3. Working Model Lineage Diagram. Genealogical relationship among the principal Working Models contributing to the current Multi-Deficit Systems Model (WM-16). Unlike Figure 2, which illustrates chronological evolution, this figure emphasizes conceptual inheritance by identifying the principal investigative branches that remained active within the present explanatory framework. WM-17 is shown separately to indicate the developmental endpoint of the current model rather than an additional explanatory branch.

Figure 3. Working Model Lineage Diagram

Editorial audit finding

The lineage reveals that the investigation has **two different kinds of evolution**, and preserving that distinction prevents the audit trail from falsely looking linear.

The **explanatory lineage** converges toward **WM-16**: increasingly specific anatomical and functional discoveries are retained as separate branches of the multi-deficit systems model.

The **developmental lineage** culminates in **WM-17**: once those components become accessible and incorporable, the question becomes whether they progress through **access** → **incorporation** → **reliability** → **automaticity**.

So the terminal architecture is not properly represented as:

WM-16 → WM-17, therefore WM-16 is obsolete.

It is:

WM-16 = current explanation of the system

WM-17 = current endpoint for the system

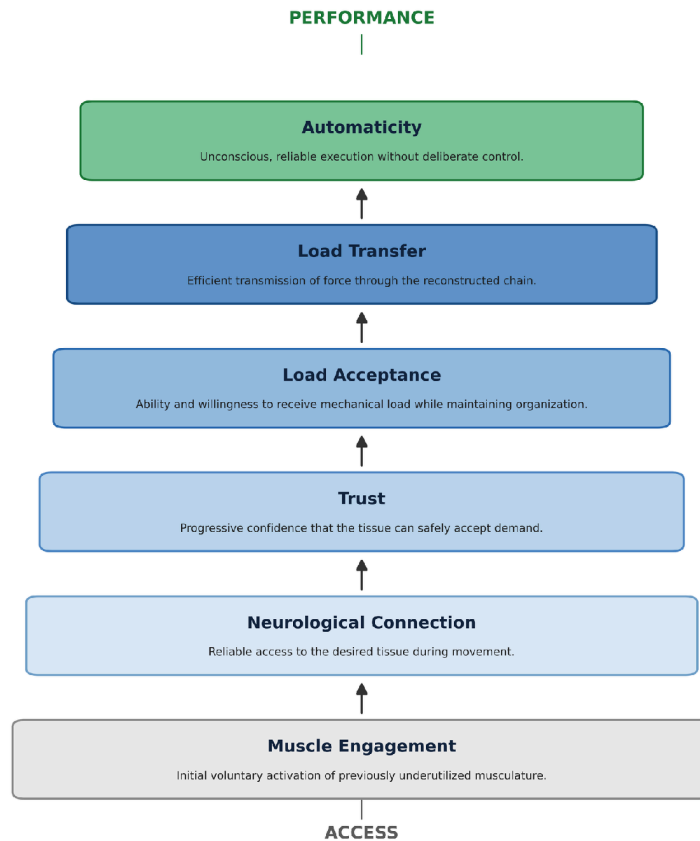
And underneath both remains the retained functional hierarchy established through WM-05/06/11:

muscle engagement → neurological connection → trust → load acceptance → load transfer.

That gives the completed section a traceable lineage without altering any of its substantive findings.

Figure 5. Functional Hierarchy

Proposed progression from tissue access to reliable performance



Note. Progression through the hierarchy is iterative rather than strictly linear. Not every movement progresses through every level during a single session. Temporary regression may occur because of fatigue, novel loading, or unresolved restriction.

Progression through this hierarchy was inferred from repeated longitudinal observations rather than from any single exercise or session.

Figure 5. Functional Hierarchy. Proposed functional hierarchy describing the progression by which previously underutilized tissues become incorporated into coordinated movement during the investigation. The hierarchy reflects the longitudinal observations documented throughout the investigation, suggesting that restoration progresses from initial voluntary muscle engagement, through increasingly reliable neurological access and confidence, toward load acceptance, effective load transfer, and ultimately automatic movement organization. The hierarchy represents a functional working model derived from repeated observation and is intended as a conceptual framework rather than a validated physiological sequence.

Figure 5. Functional Hierarchy

Chapter 6C — Current Multi-Deficit Systems Model

Status: Working hypothesis — NOT medical diagnosis

The evidence currently does **not** support a single-cause explanation.

The working structure is:

Primary functional condition:
Left chain offline / incompletely incorporated

with multiple distinguishable sub-issues:

Sub-issue	Observed functional effect	Evidentiary status
Original adductor injury/history	Persistent tightness and restricted hip flexion	Direct observation; exact anatomy unknown
Two proximal restriction centers	Restricted hip hingement/glute-pelvic movement	Direct observation; tissue identity unknown
Foot tightness/release issue	Foot/arch restriction with episodic releases/pops	Direct observation; mechanism unknown
Ankle muscular-control deficit	Impaired base bracing, ground-grabbing and downward-pressure capability	Direct observation + working interpretation
Push-off deficit	Left fails to propel/catch step like right	Direct observation
Knee/thigh stabilization deficit	Incomplete step/load stabilization	Direct observation
Glute/pelvic/trunk disengagement	Incomplete proximal load acceptance	Direct observation
Incomplete automaticity	Correct pattern increasingly accessible but still requires deliberate control	Current working interpretation

Figure 4. Current Multi-Deficit Systems Model

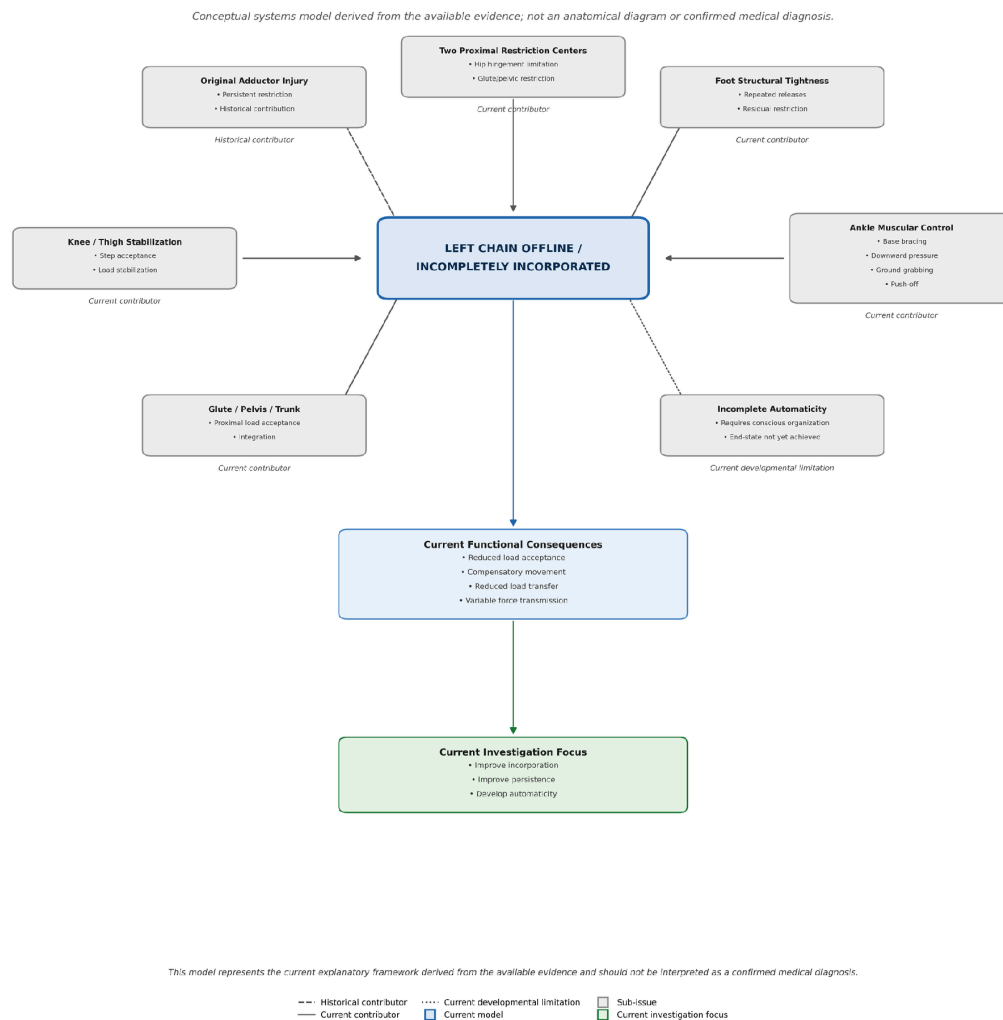


Figure 4. Current Multi-Deficit Systems Model

The critical correction established August 28 is:

The ankle is not the cause of the entire left-chain dysfunction.

Rather, it appears to be **one previously unresolved sub-issue whose particular effects help explain the inability to brace at the base, grab the ground, push downward normally and fully incorporate the distal end of the chain.**

The two proximal centers remain separate unresolved components affecting **hip hinging and proximal movement availability.**

Foot tightness/release remains another related but not yet fully explained component.

These separate problems converge functionally in the larger phenomenon you have called the **left chain being offline**.

Chapter 6D — Open Case Items & Future Investigation

These remain unresolved and should **not** be silently converted into conclusions later:

1. Exact anatomical identity of both proximal restriction centers.
2. Relationship, if any, between original adductor injury and the two centers.
3. Mechanism responsible for sitting-induced radiating numbness.
4. Structures responsible for repeated foot/ankle/hip popping and subsequent relief.
5. Exact ankle muscles/actions responsible for each portion of the perceived 360° bracing response.
6. Whether specific ankle directions reliably correspond to specific glute/trunk regions.
7. Contribution of remaining foot tightness to base-bracing limitations.
8. Whether improved ankle control changes available RDL/hinge ROM.
9. Whether the August 28 chain organization reproduces without deliberate conscious intervention.
10. Point at which deliberate incorporation becomes reliable automatic movement.
11. Whether additional unidentified sub-issues remain within the left chain.
12. Exact onset date of the original inability to apply downward pressure.
13. Exact chronology between original adductor injury and recognition of broader chain dysfunction.

Case file remains open. New evidence should amend these entries rather than overwrite earlier observations or interpretations.

SECTION 7 — SUPPORTING EVIDENCE INDEX

7.1 Purpose

This index identifies the supporting materials referenced throughout the investigation and provides a structured inventory of the documentary, photographic, medical, and observational evidence available to support the findings presented in this case file.

The purpose of the index is not to reproduce the evidence itself, but to establish a traceable record of its existence, location, and relationship to the investigative record.

Whenever possible, supporting materials should be retained in their original form to preserve evidentiary integrity. If updated versions are created for publication or presentation, the original source material should remain archived.

This index is intended to expand as additional evidence becomes available and as future versions of the case file are released.

7.2 Evidence Categories

Supporting evidence is organized into the following categories:

- Medical Documentation
- Training Documentation
- Investigative Documentation
- Photographic Evidence
- Video Evidence
- Figures & Diagrams
- Future Evidence

Each item is assigned a unique Evidence Identifier (SE-###) to facilitate cross-referencing throughout the report.

Figure 10. Supporting Evidence Architecture

Organization of the evidentiary framework supporting the current Working Model.

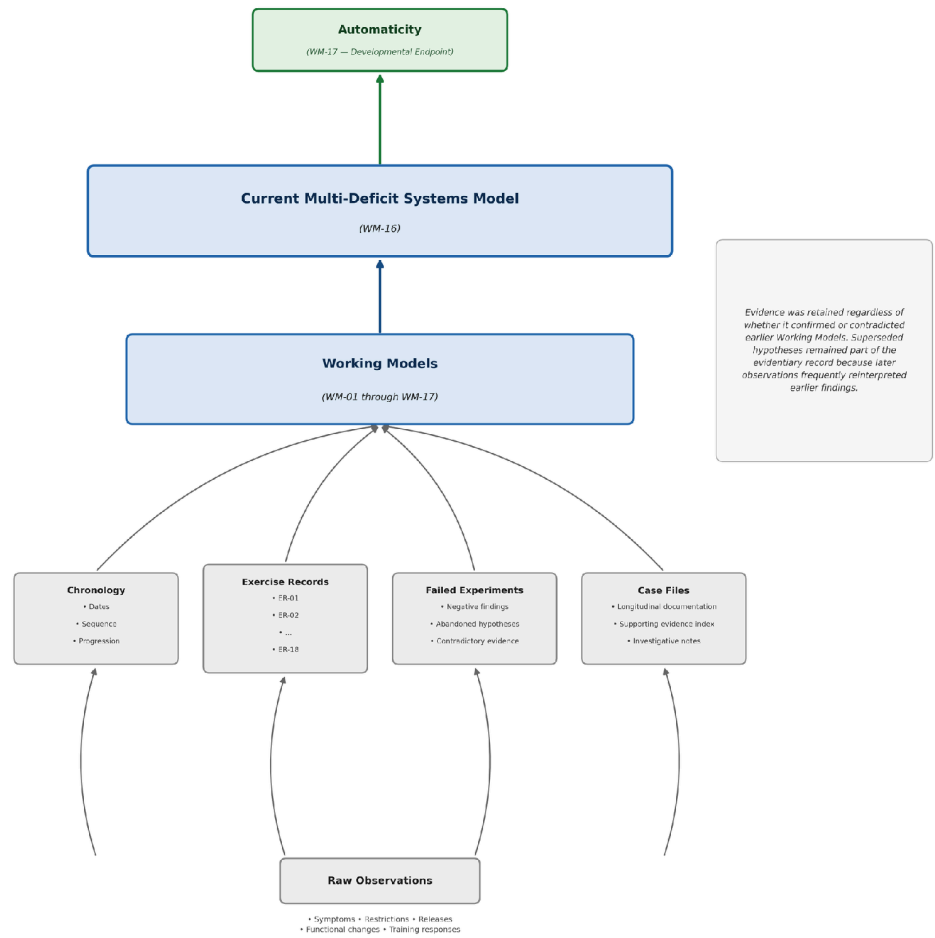


Figure 10: Supporting Evidence Architecture. Conceptual overview of the evidentiary framework underlying the present investigation. Raw observations were documented longitudinally through chronological records, exercise-specific evidence, failed experiments, and case-file documentation. These evidence sources collectively informed the progressive development of successive Working Models (WM-01 through WM-17), culminating in the current Multi-Deficit Systems Model (WM-16) and its associated developmental endpoint of Automaticity (WM-17). The figure emphasizes that the current explanatory framework emerged from converging evidence across multiple independent documentation systems rather than from any single observation or experiment.

Figure 10. Supporting Evidence Architecture

7.3 Medical Documentation

Evidence ID	Description	Status	Related Sections
SE-001	Emergency Department records relating to severe left upper-extremity swelling	Available	Chronology; CF series
SE-002	Deep Vein Thrombosis (DVT) diagnosis documentation	Available	Chronology; CF series
SE-003	Thoracic Outlet Syndrome (TOS) diagnosis documentation	Available	Chronology; CF series
SE-004	Anticoagulation treatment records (if retained)	Pending Verification	Chronology
SE-005	Additional physician records relating to upper-extremity symptoms	Pending Collection	Chronology

7.4 Training Documentation

Evidence ID	Description	Status	Related Sections
SE-101	Complete workout log documenting lower-body investigation	Available	Exercise Evidence Records
SE-102	Exercise progression history	Available	Exercise Evidence Records
SE-103	Training modifications throughout investigation	Available	Methodology; ER series
SE-104	Exercise-specific observations recorded during training	Available	ER series
SE-105	Load progression records	Available	ER series

7.5 Investigative Documentation

Evidence ID	Description	Status	Related Sections
SE-201	Complete Case File series (CF-01 through CF-20)	Available	Section 5
SE-202	Exercise Evidence Records (ER-01 through ER-18, including ER-03A and ER-09A — 20 total)	Available	Section 5
SE-203	Failed Experiments & Abandoned Hypotheses	Available	Section 6
SE-204	Evolution of the Working Model (WM-01 through WM-17)	Available	Section 6
SE-205	Current Multi-Deficit Systems Model	Available	Section 6
SE-206	Chronological reconstruction	Available	Section 4
SE-207	Methodology interview and reconstruction	Available	Section 3

7.6 Photographic Evidence (Future Supporting Materials)

The items below are not yet collected and are not part of the current version's evidentiary record.

Evidence ID	Description	Status	Related Sections
SE-301	Progress photographs documenting physical development	To Be Collected	Multiple
SE-302	Comparative left-right musculature photographs	To Be Collected	CF; ER
SE-303	Anatomical reference photographs illustrating symptomatic regions	Optional	Case Files
SE-304	Exercise setup photographs	Optional	Exercise Records

7.7 Video Evidence (Future Supporting Materials)

The items below are not yet collected and are not part of the current version's evidentiary record.

Evidence ID	Description	Status	Related Sections
SE-401	Walking analysis recordings	Planned	CF; ER
SE-402	Exercise technique recordings	Planned	Exercise Records
SE-403	Load-acceptance demonstrations	Planned	Exercise Records
SE-404	Comparative left-right movement recordings	Planned	Working Model

7.8 Figures & Diagrams

Figure	Title	Location	Status
Fig. 1	Overall Investigation Timeline	Sections 4.3–4.4	Included
Fig. 2	Evolution of the Working Model	Chapter 6B	Included
Fig. 3	Working Model Lineage Diagram	Chapter 6B	Included
Fig. 4	Current Multi-Deficit Systems Model	Chapter 6C	Included
Fig. 5	Functional Hierarchy (Muscle Engagement → Neurological Connection → Trust → Load Acceptance → Load Transfer → Automaticity)	Chapter 6B	Included
Fig. 6	Left-Chain Functional Organization	CF-02	Included
Fig. 7	Exercise Evidence Relationship Map	Section 5.6	Included
Fig. 8	Investigative Cycle	Section 3.3	Included
Fig. 9	Exercise Classification Map	Section 5.5	Included

Figure	Title	Location	Status
Fig. 10	Supporting Evidence Architecture	Sections 7.1-7.2	Included

7.9 Cross-Reference Matrix

Supporting evidence frequently contributes to multiple portions of the investigation.

Accordingly:

- Medical documentation primarily supports the Chronology and relevant Case Files.
- Training documentation supports both the Exercise Evidence Records and the evolution of the Working Model.
- Photographs and videos provide visual corroboration of observations documented elsewhere.
- Figures and diagrams serve explanatory rather than evidentiary purposes and are intended to improve communication of complex investigative relationships.

Readers are encouraged to interpret individual evidence items within the broader context of the complete investigative record rather than as isolated proof of any single conclusion.

7.10 Evidence Integrity

Whenever possible, supporting evidence should be retained in its original format.

If edited copies are created for publication, presentation, or educational purposes, the original materials should remain archived to preserve authenticity and permit future verification.

Future versions of this case file should expand this index rather than replacing existing entries, thereby maintaining a continuous evidentiary chain throughout the life of the investigation.

7.11 Current Evidence Inventory (Version 1.0)

As of **Case File Version 1.0 (Record Through August 28, 2026)**, the investigation includes:

Evidence Type	Quantity
Case Files	20
Exercise Evidence Records	20
Working Models	17

Evidence Type	Quantity
Failed Experiments / Abandoned Hypotheses	21
Chronological Records	1 Comprehensive Timeline
Methodology Reconstruction	1
Workout Logs	1 Complete Log
Medical Diagnoses	2 Confirmed (DVT, TOS)
Figures & Diagrams (Included)	10
Future Video Series (Not Yet Collected)	4 Categories
Future Photographic Series (Not Yet Collected)	4 Categories

This index is intended to function as the documentary bridge between the narrative of the investigation and its supporting materials, enabling future readers, reviewers, clinicians, and researchers to locate, verify, and interpret the evidence upon which the investigation is based.

SECTION 8 — APPENDICES

8.1 Purpose

The appendices contain supplementary reference material supporting the investigation but not forming part of the primary evidentiary record or analytical discussion.

These materials are provided to improve clarity, facilitate future revision, maintain consistency across versions of the case file, and assist readers in interpreting the investigation. Whenever possible, appendices are maintained as living reference documents that may expand in future editions without altering the historical content of the main report.

Unless otherwise indicated, the appendices are editorial reference material and should not be interpreted as independent evidence.

8.2 Appendix Organization

The appendices are organized into the following reference categories:

Appendix	Title	Purpose
Appendix A	Glossary of Investigative Terminology	Defines technical and investigation-specific terms used throughout the report.
Appendix B	Abbreviations & Acronyms	Provides standardized abbreviations used within the case file.

Appendix	Title	Purpose
Appendix C	Evidence Classification Reference	Summarizes the evidence-classification framework and exercise evidence-strength classifications.
Appendix D	Case File Cross-Reference Matrix	Maps relationships among Case Files, Exercise Records, Working Models, and analytical sections.
Appendix E	Working Model Lineage Diagram	Graphical representation of the evolution of the Working Models.
Appendix F	Figure & Diagram Index	Lists all figures, diagrams, and illustrations included in current and future versions.
Appendix G	Revision History	Documents changes made between published versions of the case file.

8.3 Appendix A — Glossary of Investigative Terminology

This appendix defines specialized terminology developed or consistently used throughout the investigation.

Terms are included to ensure consistent interpretation and should be referenced whenever terminology is ambiguous or used in a specialized investigative context.

Examples include:

- Load Acceptance
- Load Transfer
- Reincorporation
- Working Model
- Priming
- Integration
- Automaticity
- Downward Pressure
- Base Bracing
- Left Chain
- Distal

- Proximal

Additional terms may be added in future revisions as the investigation expands.

8.4 Appendix B — Abbreviations & Acronyms

Standard abbreviations used throughout the report include:

Abbreviation	Meaning
CF	Case File
ER	Exercise Evidence Record
WM	Working Model
FE	Failed Experiment / Abandoned Hypothesis
SE	Supporting Evidence
DVT	Deep Vein Thrombosis
TOS	Thoracic Outlet Syndrome
ROM	Range of Motion
RDL	Romanian Deadlift

Future abbreviations should be added here before being introduced into the main text.

8.5 Appendix C — Evidence Classification Reference

This appendix consolidates the classification systems used throughout the report.

Evidence Classifications

- Confirmed Medical Diagnosis
- Direct Observation
- Working Interpretation / Working Hypothesis
- Unknown

Exercise Evidence Strength Classifications

- Primary Discovery Exercise
- Major Supporting Exercise
- Confirmatory Exercise
- Integration Exercise
- Priming Exercise

- Development Exercise
- Assessment Exercise

Confidence Scale

Confidence ratings used throughout this document follow a five-point scale: **High**, **Medium-High**, **Medium**, **Medium-Low**, and **Low**.

Where a compound value appears (e.g., “Medium-High”), it should be read as an intermediate level between the two named points on the scale, rather than as a separate, sixth category.

Each classification should be interpreted according to the definitions established in the front matter.

8.6 Appendix D — Case File Cross-Reference Matrix

This appendix provides a consolidated reference showing the explicit cross-references recorded within the primary evidentiary record. Relationships shown here reproduce the “Related events” and “Model Lineage” fields documented in Section 5 (Part I) and Chapter 6B; they are not inferred connections.

Case File (CF) Cross-Reference Matrix

Case File	Title	Related Case Files	Figure
CF-01	Pre-existing Left Adductor Injury	CF-02, CF-03, CF-10, CF-16	—
CF-02	Development/Re cognition of Global Left-Chain Dysfunction	All subsequent entries	Fig. 6
CF-03	Two Centers of Hip/Glute Restriction Identified	CF-04, CF-05, CF-07, CF-17	—
CF-04	Sitting-Induced Neurological Symptoms	CF-03	—
CF-05	Early Proximal Release Event / Glute Access	CF-06, CF-07	—
CF-06	Distal Foot/Ankle	CF-05, CF-09, CF-18	—

Case File	Title Release Sequence	Related Case Files	Figure
CF-07	Priming Procedure Emerges	CF-08 through CF-18	—
CF-08	Pallof Press and Hip-Airplane Findings	CF-09, CF-11, CF-18	—
CF-09	Arch Begins to “Own the Ground”	CF-06, CF-11, CF-18	—
CF-10	Adductor/Hip Release and Transition Toward Strengthening	CF-01, CF-03	—
CF-11	Knee/Thigh Catch and Push-Off Deficit	CF-09, CF-12, CF-18	—
CF-12	Single-Leg Weight-Acceptan ce Drill	CF-11, CF-14, CF-18	—
CF-13	Frog Bridge / Pelvic Integration Discovery	CF-14, CF-15	—
CF-14	Supermans / Swimmers Before Hamstring Curl	CF-17	—
CF-15	Horizontal vs. Linear Leg-Press Priming	CF-16, CF-18	—
CF-16	Bulgarian Split Squat Integration / Adductor Pinch	CF-01, CF-15, CF-18, CF-19	—
CF-17	Proximal Hamstring	CF-03, CF-14	—

Case File	Title	Related Case Files	Figure
	"Strings" Identified Under Load		
CF-18	Multidirectional Ankle-Control Discovery	CF-06, CF-09, CF-11, CF-15, CF-19	—
CF-19	Retrospective Reinterpretation of "Cannot Apply Downward Pressure"	CF-02, CF-06, CF-09, CF-11, CF-18	—
CF-20	August 28 Integrated Lower-Body Session	CF-15 through CF-19	—

Working Model (WM) Lineage Matrix

Working Model	Title	Current Descendant(s)
WM-01	Local Injury / Adductor-Centered Model	WM-16
WM-02	Glute/Hip Restriction Becomes the Dominant Proximal Model	WM-16
WM-03	Two-Center Proximal Restriction Model	WM-16
WM-04	Release Appears to Be the Key to Restoration	WM-16
WM-05	Individual Muscle Re-Engagement Model	WM-16 and WM-17
WM-06	Load Acceptance Emerges as a Central Concept	WM-16 and WM-17
WM-07	Downward Pressure and Push-Off Become Distinct Unresolved Functions	WM-15 and WM-16
WM-08	Foot/Arch Becomes Part of the Chain	WM-16

Working Model	Title	Current Descendant(s)
WM-09	Knee/Thigh Stabilization and the Missing “Catch”	WM-16
WM-10	Pelvis and Trunk Enter the Model as Active Load-Bearing Components	WM-16 and WM-17
WM-11	Load Transfer Becomes Distinct from Load Acceptance	WM-16 and WM-17
WM-12	Priming Becomes Selective Rather Than Generic	WM-16
WM-13	Horizontal Leg Press Opens the Ankle Investigation	WM-14, WM-15 and WM-16
WM-14	Multidirectional Ankle-Control Model	WM-16 and WM-17
WM-15	Retrospective Reinterpretation of the Downward-Pressure Problem	WM-16
WM-16	Current Multi-Deficit Systems Model Emerges	WM-16 and WM-17
WM-17	Automaticity Becomes the Endpoint	WM-17

A fully detailed version of the Working Model lineage — including direct parent, direct successor, final disposition, and present status for each model — appears in the Working Model Evolution Map in Chapter 6B and in Figure 3.

Exercise Evidence Records (ER) and Failed Experiments (FE) are not cross-referenced by identifier elsewhere in the case file and are therefore not included as matrix columns in this version. Their relationships to specific Case Files and Working Models are described in narrative form within their respective entries in Section 5 and Chapter 6A.

8.7 Appendix E — Working Model Lineage Diagram

The graphical lineage of the Working Models developed throughout the investigation is embedded as Figure 3 in Chapter 6B, rather than presented as separate content in this appendix.

The diagram complements the written Working Model Evolution by illustrating:

- model expansion;
- refinement;
- convergence;
- partial retention;
- and integration into the current multi-deficit systems model.

8.8 Appendix F — Figure & Diagram Index

The master list of all figures and diagrams referenced throughout the report is maintained in Section 7.8 rather than duplicated here.

All 10 figures were released as part of Version 1.0 on August 28, 2026.

Future editions of the case file should update Section 7.8 as new illustrations are added.

8.9 Appendix G — Revision History

To preserve the historical integrity of the investigation, revisions to the case file should be documented rather than silently incorporated.

Each published version should record:

Version	Date	Summary of Changes
1.0	August 28, 2026	Initial published Case File, including the completed figure package.
Future	—	Subsequent revisions to be documented upon release.

The purpose of this appendix is to maintain a transparent editorial history and allow readers to distinguish between newly added evidence, refined interpretations, and structural revisions.

Closing Statement

This case file represents the first complete reconstruction of a longitudinal self-case investigation into persistent left-chain dysfunction, movement reincorporation, and load-transfer restoration.

The investigation remains active. Accordingly, the findings presented herein should be interpreted as the current state of the evidentiary record rather than as final conclusions.

Future versions will preserve the historical progression of the investigation while incorporating new observations, supporting evidence, and refinements to the working model. In doing so, the document is intended to function not only as a record of one individual's investigation, but also as a transparent example of evidence-driven inquiry, where observations are preserved, uncertainty is acknowledged, and understanding advances through disciplined investigation rather than assumption.